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Paleomagnetism of the southwestern U.S.A. recorded by 0–5 Ma igneous rocks

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[1] The issue of permanent nondipole contributions to the time-averaged field lies at the very heart of paleomagnetism and the study of the ancient geomagnetic field. In this paper we focus on paleomagnetic directional results from igneous rocks of the southwestern U.S.A. in the age range 0–5 Ma and investigate both the time-averaged field and its variability about the mean value. Several decades of work in the southwestern United States have resulted in the publication of paleomagnetic data from over 800 individual paleomagnetic sites. As part of a new investigation of the San Francisco Volcanics, we collected paleomagnetic samples from 47 lava flows, many of which have been previously dated. The new data combined with published data are highly scattered. Contributions to the scatter were considered, and we find that removal of data sets from tectonically active areas and judicious selection according to *Fisher*'s [1953] precision parameter results in an axially symmetric data distribution with normal and reverse modes that are indistinguishable from antipodal. Monte Carlo simulations suggest that a minimum of 5 samples per site are needed to estimate the precision parameter sufficiently accurately to allow its use as a determinant of data quality. Numerical simulations from statistical paleosecular variation models indicate the need for several hundred paleomagnetic sites to get an accurate determination of the average field direction and are also used to investigate the directional bias that results from averaging unit vectors rather than using the full field vector. Average directions for the southwestern U.S.A. show small deviations from a geocentric axial dipole field, but these cannot be considered statistically significant. Virtual geomagnetic pole (VGP) dispersions are consistent with those from globally distributed observations analyzed by *McElhinny and McFadden* [1997]. However, a systematic investigation of the effect of imposing a cutoff on VGPs with large deviations from the geographic axis indicates that while it may reduce bias in calculating the average direction, such a procedure can result in severe underestimates of the variance in the geomagnetic field. A more satisfactory solution would be to use an unbiased technique for joint estimation of the mean direction and variance of the field distribution.

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Theme: Geomagnetic Field Behavior Over the Past 5 Myr
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1. Introduction

[2] Over the past four decades the key assumption in many paleomagnetic studies has been that the average direction of the paleomagnetic field corresponds to one that would have been produced by a geocentric axial dipole (GAD), yet most studies of the time averaged field have suggested significant departures from GAD. The issue of permanent nondipole and even nonzonal contributions to the time averaged field lies at the very heart of paleomagnetism and the study of the ancient geomagnetic field. As simulations of the geomagnetic field improve, the problem of determining the nature of the real geomagnetic field from observations gains urgency.

[3] Recently produced models for the time averaged geomagnetic field (TAF) for the last five million years have made use of compilations of paleomagnetic field observations derived from paleosecular variation (PSV) studies on lava flows [e.g., *Gubbins and Kelly*, 1993; *Kelly and Gubbins*, 1997; *Quidelleur et al.*, 1994; *Johnson and Constable*, 1995, 1996, 1997, 1998; *McElhinny and McFadden*, 1997; *Carlut and Courtillot*, 1998; *Carlut et al.*, 2000]. Such models require data from well-distributed sites over the entire globe that represent both polarities and meet a consistent set of minimum standards for data quality.

[4] Because procedures for sampling, demagnetizing and measuring have changed over four decades of intensive paleomagnetic research, standards of data quality have evolved to the point where one might question whether older data sets are still useful for the purposes of TAF modeling. *McElhinny and McFadden* [1997] (hereinafter MM97) drew attention to this issue and recommended that a number of older studies be redone. To address the need for high quality data, paleomagnetists have begun the process of augmenting the data on which the TAF models are based. [see, e.g., *Johnson et al.*, 1998; *Carlut et al.*, 2000; *Tauxe et al.*, 2000; *Camps et al.*, 2001]. In this paper we present new results from Plio-Pleistocene rocks of the San Francisco Volcanics and reanalyze existing published data from the southwestern United States.

2. Previous Work in the Southwestern United States

[5] There have been eleven studies published with fully oriented paleomagnetic data from the southwestern United States (see Tables 1 and 2 and Figure 1). Selected data from these studies were included in the PSVRL database (see MM97).

[6] Because varying criteria for selecting data may lead to different conclusions, we have taken the approach of compiling all the data published, which give the precision and/or angular cones of 95% confidence about the mean direction, regardless of any perceived quality factors such as number of samples per site, or the Fisherian parameters. In addition we keep all directions that might be categorized as intermediate, in order to avoid making arbitrary judgements about what constitutes a transitional or excursional direction. Our compilation includes the unit identifier, the mode of emplacement where known (flow, vent, dike, dome, tuff, etc.), the site location in latitude and longitude, age, paleomagnetic direction, the number of lines and great circles used to constrain the average direction, and *Fisher*'s [1953] parameters designated α_{95}, and k.

[7] As part of a new paleomagnetic database effort, we have developed a set of metadata and a template for data entry. These are described in detail on the Web site http://earthref.org/cgi-bin/erda.cgi?n=135. All of the published data compiled for this paper are available online at http://earthref.org/cgi-bin/err.cgi?n=1193 as well as the new data to be described later. These are plotted in equal area projection in Figures 2 and 3. We will briefly outline the various studies from west to east.

Table 1. Publications With Paleomagnetic Data From the SW United States[a]

Map Number	PSVRL00	Lat., N	Long., E	GAD *I*
1*	100	38.5	−122.5	57.8
2*	95	39.5	−120	58.7
3	97	37.7	−119	57.1
4	123	35.4	−112	54.9
5	94	35.2	−111.6	54.7
6	–	35.5	−107	55.0
7	98	35.9	−106.5	55.4
8*	101	36.6	−105.6	56.0
9	120	∼36.7	∼−112.5	∼56.1

[a] Locations are shown in Figure 1. Studies marked with an asterisk are tectonically suspect. 1, *Mankinen* [1972, 1989], mankinen72_89.xls; m89.xls; 2, *Heinrichs* [1967], heinrichs67.xls; 3, *Mankinen et al.* [1986], mankinen86.xls; 4, *Tanaka et al.* [1990], tanaka90.xls; 5, *Kono et al.* [1967], kono67.xls; 6, *Geissman et al.* [1990], geissman90.xls; 7, *Doell et al.* [1968] and *Geissman* [1988], doell68.xls; 8, *Brown et al.* [1993], brown93.xls; 9, *Champion* [1980], champion80.xls. These data files use the metadata described in PRMAG.metadata.v11.AA#8FB20.xls.

[8] The studies farthest to the west [*Mankinen*, 1972, 1989] (#1 in Figure 1 and Tables 1 and 2) were on the Sonoma Volcanics of Northern California. We have combined these two studies together because the latter was a reanalysis of the former by the same author. A total of 43 cooling units were sampled ranging in age from about 8 Ma to about 3 Ma; most of the sites are between 5.5 and 4 Ma. The stability of remanence was tested on at least one specimen per site by step-wise alternating field demagnetization. The rest of the specimens were treated to demagnetization at a single field step, a procedure often termed "blanket demagnetization." Pilot specimens from several flows showed little change during demagnetization and the NRM vectors were well-grouped. For these, the NRM directions were used.

[9] Most of the volcanic units sampled in the Sonoma Volcanics required tilt "correction." Sedimentary bedding was used to approximate structural attitude although this was difficult to determine accurately in many cases. The resulting tilt adjusted data set (shown in Figure 2a), is quite scattered; both the normal and reverse modes are distinct from the expected GAD direction. *Mankinen* [1989] speculated that the cause of the elliptical distribution in these directions was incomplete averaging of geomagnetic secular variation. We find it likely that inappropriate tilt adjustments contributed to the scatter in the Sonoma volcanics, a "positive" fold test notwithstanding. The possibility of tectonic complications in these data is indicated in Table 1.

[10] The study carried out at #2 in Figure 1 was on the Plio-Pleistocene Lousetown Formation near Lake Tahoe. *Heinrichs* [1967] sampled 61 lava flows in three fault bounded regions: the Virginia range, the Carson range and the Truckee basin.

Table 2. Summary Statistics for the Studies in Table 1[a]

Study	Mode	$\bar{D}$	$\bar{I}$	α_{95}	*N*	*k*	Fisherian?	GAD?
1	N	349.2	43.7	14.5	15	8.	yes	no
1	R	181.1	−47.3	6.6	28	18.	yes	no
2	N	27.1	52.6	17.4	10	9.	yes	yes
2	R	135.3	−76.4	9.1	51	6.	no	no
3	N	0.6	58.6	4.5	39	27.	yes	yes
4	N	356.7	52.5	2.5	250	14.	no	no
4	R	181.7	−52.4	2.1	311	15.	no	no
5	N	11.5	47.1	24.5	3	26.	yes	yes
5	R	173.4	−49.6	9.0	19	15.	yes	yes
6	?	101.1	−36.1	1.2	8	2150.	yes	no
7	N	4.2	51.8	6.2	8	81.	yes	yes
7	R	180.6	−55.8	14.5	8	16.	yes	yes
8	N	0.9	44.7	5.8	28	23.	no	no
8	R	166.6	−49.1	9.1	32	9.	no	no
9	N	3.6	53.7	8.9	10	31.	no	yes
Tectonically	N	[324.7]	[87.6]	2.1	310	15	no	yes
undisturbed	R	[169.4]	[−87.8]	2.2	346	13	no	yes
Tectonically	N	[6.4]	[79.4]	6.0	53	11	no	no
disturbed	R	[55.6]	[−81.8]	5.8	111	6	no	no

[a] Numbers in brackets are in D', I' coordinates (see text).

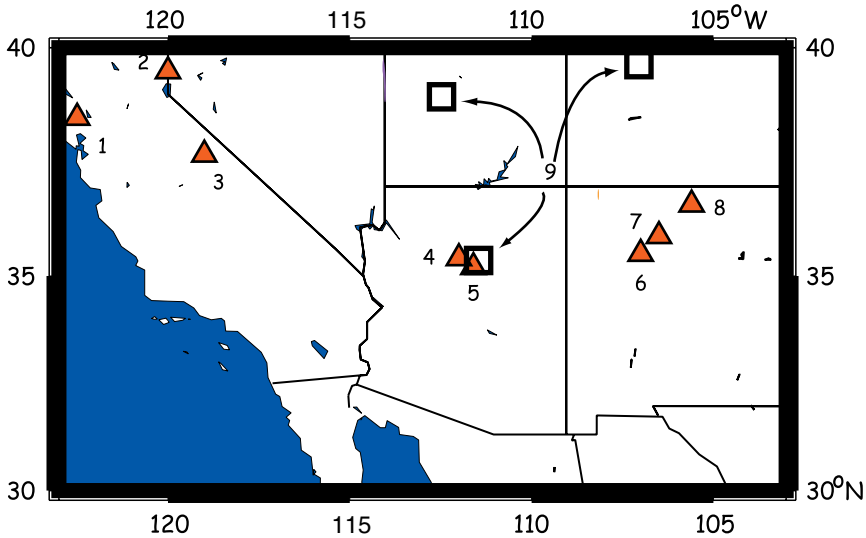


Figure 1. Map of southwestern United States showing location of previous studies. See Table 1. Squares are from *Champion* [1980].

[11] The group comprising sites 1–43 were taken from the Virginia range and have an estimated age of 6.8 Ma or older. Sites 3–33 (directions enclosed by the dotted line on Figure 2b) come from Lousetown Creek in the Virginia Range. These may be transitional between the normal (site number 34) and reverse (site number 1) bounding sites. The Lousetown Creek units were reanalyzed [*Roberts and Shaw*, 1990] who found that several flows had substantial overprints presumed to be from lightning strikes. No new site means were presented. We find it hard to imagine that the widespread coherent direction represented by multiple flows contained within the dashed line in Figure 2b could be the result of lightning strikes, which tends to result in highly scattered data. Nonetheless, lightning is likely to have played a role in some sites with poor within site reproducibility. In any case, these sites are likely to be older than the 5 Ma maximum age that we designated as a cutoff in the time averaged field project.

[12] The second group of sites, 44–53 and 62–64 were collected from the Carson Range. There is no age control on these units. Many of the sites are just east of a major fault which forms the eastern edge of the Truckee basin. The third group of sites is located within the Truckee basin. The flows from

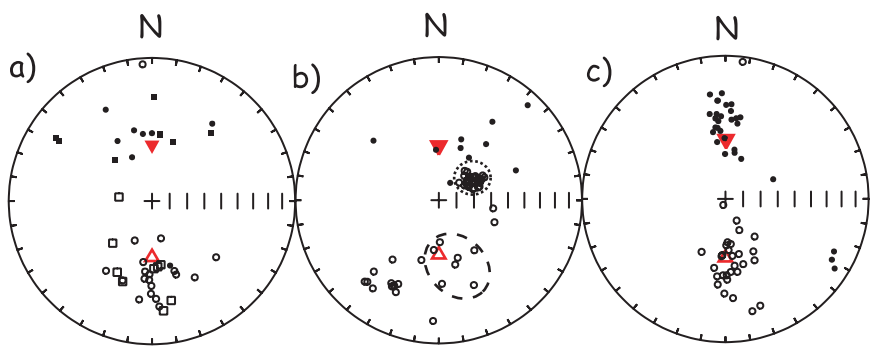


Figure 2. Equal area projections of data from studies cited in Table 1 as potentially tectonically disturbed. Solid (open) symbols are for the lower (upper) hemisphere. The triangles are the normal and reverse directions expected from a geocentric axial dipole field at the location. a) Squares are the data from *Mankinen* [1972] not used by *Mankinen* [1989]. Circles are the data from *Mankinen* [1989]; #1 in Figure 1. b) *Heinrichs* [1967]; #2 in Figure 1. Data within the dashed line are the least likely on the basis of field evidence to have been tectonically rotated. Data in the dotted circle are from Lousetown creek. See text for explanation. c) *Brown et al.* [1993]; #8 in Figure 1.

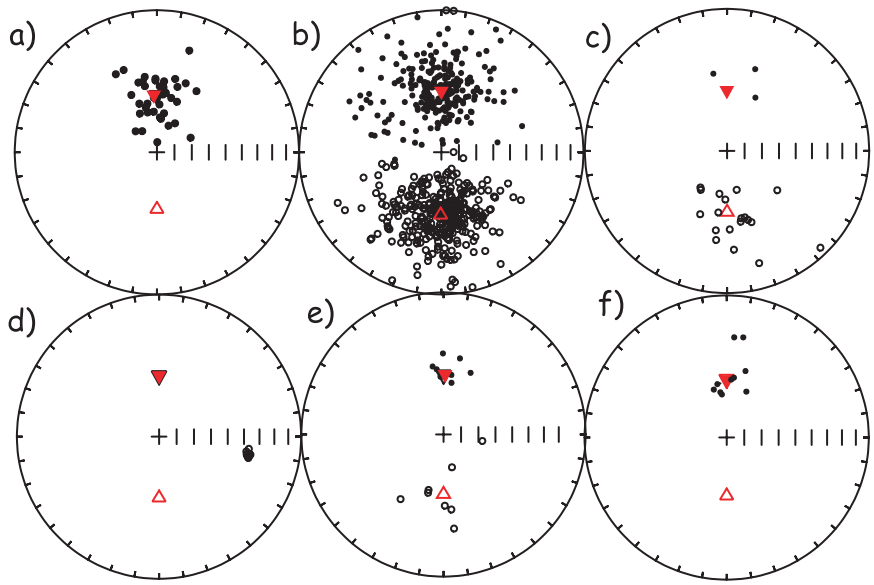


Figure 3. Equal area projections of all data from studies in tectonically stable areas cited in Table 1. Symbols same as in Figure 2. a) *Mankinen et al.* [1986]; #3 in Figure 1. b) *Tanaka et al.* [1990]; #4 in Figure 1. c) *Kono et al.* [1967]; #5 in Figure 1. d) *Geissman et al.* [1990]; #6 in Figure 1. e) *Doell et al.* [1968] and *Geissman* [1988]; #7 in Figure 1. f) *Champion* [1980]; #9 in Figure 1.

the Truckee basin (dated at approximately 1.2–1.9) are the youngest and least likely to have been tilted. Truckee basin sites yield the directions enclosed in the dashed line in Figure 2b, and are quite close to the expected reverse direction (indicated by a red triangle).

[13] At least two specimens were subjected to step-wise alternating field demagnetization from each site with the remaining specimens generally being treated at a single step. We plot the directions in Figure 2b. The reverse data are not Fisherian and are significantly different from the expected GAD direction (Table 2).

[14] The Lake Tahoe area has been extensively faulted with "several hundred feet of structural relief" noted in the Lousetown formation. Nearly all of the "deviant" directions come from regions quite close to mapped faults and it is likely that undetected rotations contribute to the unusual directions observed in this data set. We have indicated the likelihood of tectonic complication in Table 1.

[15] Study #3 [*Mankinen et al.*, 1986] investigated samples from the Long Valley Caldera, California. *Mankinen et al.* [1986] published data for a total of 39 basalts, trachyandesites, quartz latite domes, and rhyolite domes and flows. All of the sites are of normal polarity as shown in Figure 3a and are consistent with the expected direction from a GAD field (Table 2). Most of the sites have K-Ar ages and the data span the age interval 0.064–0.751 Ma. From three to eight oriented cores were taken from each site and an attempt was made to cover a sufficiently wide outcrop span to detect possible postcooling rotation, particularly in the rhyolite domes. At least one specimen from each site was subjected to step-wise alternating field (AF) demagnetization experiments. Fourteen site means use the NRMs, and the rest are from blanket demagnetizations at steps ranging from 5 to 70 mT, based on the behavior of the pilot specimens.

[16] *Tanaka et al.* [1990] (#4 in Figure 1) compiled data for 580 lava flows, dikes, domes and vents from the San Francisco Volcanics in and around Flagstaff, Arizona. Many K-Ar dates and a few ^{14}C dates are available for these sampling sites which range in age from less than 1 ka to about 4.5 Ma. The paleomagnetic data were collected over a period of nearly three decades by many investigators, most of whom were interested in the polarity of the lava flows and not necessarily in precise directions. Many sites were taken as oriented block

samples that were later drilled in the laboratory. While some sites had samples that were spread over many meters, most were sampled over a rather short range (1–2 meters). Some specimens were subjected to careful, step-wise alternating field, while others received blanket, or no demagnetization treatment at all. There are several data points given by *Tanaka et al.* [1990] that were included from another published study in the area [*Champion*, 1980] which we will discuss later; these have been removed from Tanaka et al.'s data table to avoid duplication. *Tanaka et al.*'s [1990] data are shown in Figure 3b. Because of the large number of sites, the α_{95}s of the normal and reverse mean directions are quite small ($\sim 2^\circ$) and exclude the expected direction from a GAD field. The distribution of site means is non-Fisherian (see Table 2).

[17] *Kono et al.* [1967] (#5 in Figure 1 and Tables 1 and 2) published data from 22 basaltic lava flows of which three were normal, one was intermediate and the rest were reverse. These data are from two sections, one near #5 on the map shown in Figure 1 and one near Taos, New Mexico. The lava flows are separated by sands and gravels so may represent temporally distinct recordings of the geomagnetic field. The stability of the NRM was tested with baked contact tests, storage and low-temperature demagnetization tests, as well as AF and thermal demagnetization. Generally, two specimens from each lava flow were subjected to step-wise AF demagnetization. At least one specimen from each lava flow was subjected to step-wise thermal demagnetization. We have listed all data considered "stable" by the authors. The normal and reverse modes are both Fisherian and consistent with the direction expected from a GAD field (Figure 3c, Table 2).

[18] *Geissman et al.* [1990] (#6 in Figure 1 and Tables 1 and 2) published data from eight basaltic lava flows near Albuquerque, New Mexico. The dates in that paper were superseded by ^{238}U-^{230}Th dates given by *Peate et al.* [1996]. The paleomagnetic vectors are indistinguishable among these sites and there are no weathering horizons between the lava flows. It is thus reasonable to suppose that these were very rapidly extruded and may constitute a "spot" reading of the geomagnetic field. At least seven separately oriented samples were taken over a wide outcrop span from each site. Cores were oriented with both magnetic and sun compass measurements. Specimens were subjected to either AF or thermal demagnetization. About half displayed simple single component demagnetization behavior, while the rest demagnetized along great circles. Most of the directions recovered are apparently based on principal component analysis [*Kirschvink*, 1980], but some are derived from combinations of best-fit lines and planes [*Halls*, 1978; *Bailey and Halls*, 1984]. The tightly grouped directions (Figure 3d) are quite deviant from the expected GAD field (Table 2).

[19] *Doell et al.* [1968] (#7 in Figure 1 and Tables 1 and 2) published data from the Valles Caldera near Taos, New Mexico. They reported directions from 16 ash flow tuffs and rhyolite domes (see Figure 3e). The mean directions are consistent with those expected from a GAD field. There are K-Ar age data for most of the sites. Two of these cooling units (the South Mountain Rhyolite and the Battleship Rock Tuff) were resampled by *Geissman* [1988], who quoted nearly identical values; we have removed the superseded sites from the compilation described here. From 5 to 12 separately oriented cores were taken from each site, spanning lateral distances of tens of meters up to as much as 8 km. Pilot specimens were subjected to step-wise AF demagnetization and the rest of the specimens from each site were treated by blanket demagnetization from 20 to 40 mT. There is an allusion to the acquisition of ARM at higher peak fields, so the level of blanket demagnetization was chosen to balance the elimination of secondary magnetization (presumably lightning-induced IRM) and the acquisition of anhysteretic remanent magnetization (ARM) at higher peak fields. Directions from two sites are based on the NRM. Thermal demagnetization was found to be largely ineffective in removing the secondary (lightning) magnetization.

[20] *Brown et al.* [1993] (#8 in Figure 1 and Tables 1 and 2) reported data from the Taos Plateau

volcanic field, Rio Grande rift, New Mexico. They collected data from 60 basaltic and andesitic flows (29 reverse, 4 intermediate, 27 normal) ranging in age from 1.8 to 4.7 Ma. The directions from this study are shown in Figure 2c. Sites 40–51 and sites 52–59 are part of two stratigraphic sequences. Sediments are interspersed in these sections but temporal independence is questionable for sites 40–43, 44–47, 48–49, 50–51, 52–53, 54–55(A and B) and 56–58. Five to 14 separately oriented samples were taken from each lava flow and orientated using a sun compass. Pairs of specimens were subjected to step-wise AF and thermal demagnetization. Best-fit lines and planes were calculated for the demagnetization data using the method of *Kirschvink* [1980] and combined to give site mean directions and confidence intervals. The paper considers the possibility of some tilting of the region and concludes that the southern and eastern parts of the plateau could have been significantly (~8°) downwarped contributing to the observed inclination anomaly of the average direction (Table 2).

[21] *Champion* [1980] (#9 in Figure 1 and Tables 1 and 2) presented data from a number of paleomagnetic sites from the region of interest here. He sampled Holocene vents, flows and cinder cones near Flagstaff and lava flows from several other localities (see Figure 1). All samples were drilled, and at least six separately oriented samples were taken from two separate localities in each cooling unit. The data from the two localities were combined to give cooling unit averages. One specimen from each locality was subjected to step-wise AF demagnetization. Because neither specimen from the two localities within each cooling unit showed any change during demagnetization, many of the cooling unit averages used here were based on the NRM directions. These data, plotted in Figure 3f, are consistent with a GAD direction (Table 2).

[22] The data shown in Figures 2 and 3 are quite scattered as might be expected from the absence of any selection criteria and there are a number of reasons to proceed with caution. Some parts of southwestern U.S.A. suffer from severe and frequent lightning storms [see, e.g., *Krider and Roble*, 1986; *Verrier and Rochette*, 2002]. Lightning has long been a concern for paleomagnetists [e.g., *Graham*, 1961], resulting in deleterious effects on the paleomagnetic record that may not be adequately compensated for in subsequent cleaning and analyses. Because of the severity of lightning and the preponderance of highly scattered data in the southwestern U.S. data sets, it would be interesting to know if more detailed field and laboratory procedures and some form of selection criteria would substantially alter the picture of the time averaged field and paleosecular variation in the southwestern United States. To investigate this, we have conducted a new field and laboratory effort in the San Francisco Volcanic region of northern Arizona.

3. New Paleomagnetic Data From the San Francisco Volcanics

[23] Drawing on the substantial information concerning outcrop availability, likely success in obtaining reliable results and a wealth of age information in the *Tanaka et al.* [1990] (hereinafter TOS90) publication, we revisited the San Francisco Volcanics during two field trips conducted in 1999 and 2000.

[24] Figure 4 shows the geology of the area with our sampling locations (Tables 3 and 4). The San Francisco Volcanic Field is a Plio/Pleistocene volcanic province with ages ranging from very recent (some 925 years ago) to some 5 million years old [*Wolfe et al.*, 1983]. The geology was described by TOS90 and references therein. The San Francisco volcanic (SV) province is bordered on the south by the Tertiary "rim basalts" which are mostly Miocene in age. There is a general trend in ages from the oldest in the west (~4 Ma) to the youngest in the east (less than 1 ka). Rocks range in lithology from basalt to andesite and occur as flows, vents and domes.

[25] In our first field season (August, 1999), we attempted to relocate the isotopically dated lava flows sampled by TOS90. Although we found a number of sites whose TOS90 identity were unambiguous, the absence of high precision site locations meant that many TOS90 target sites were

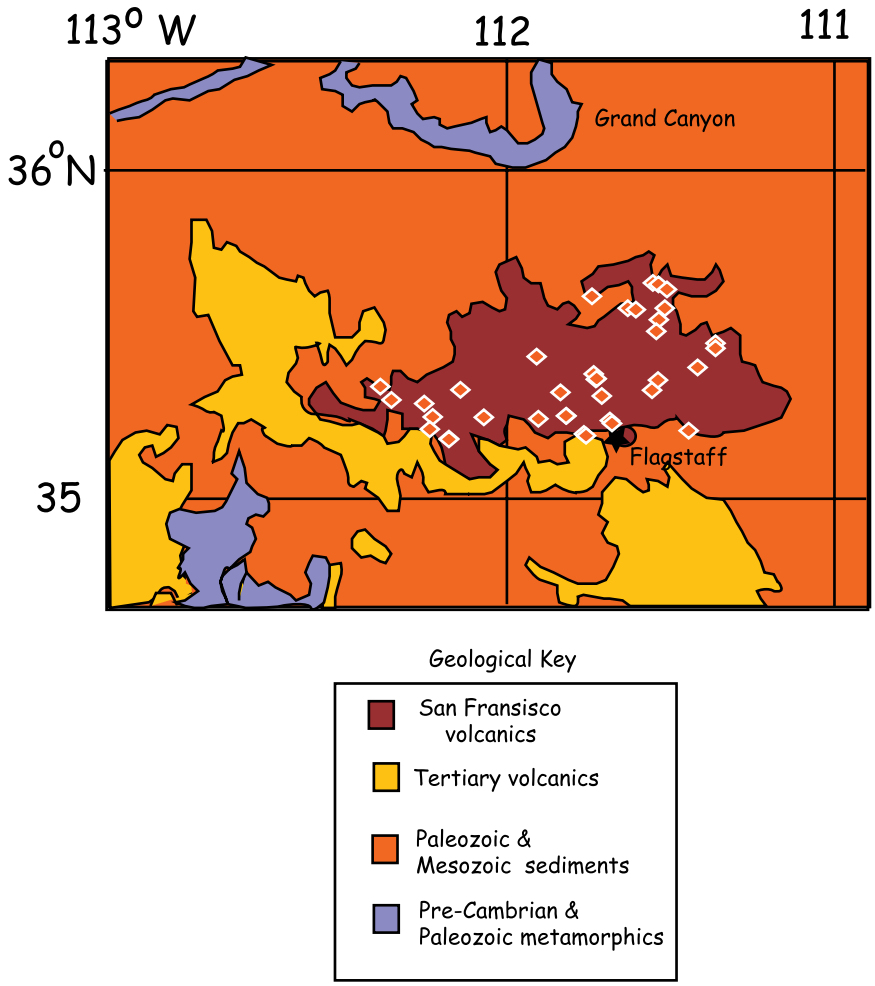


Figure 4. Map showing the geology and sampling sites of the present study. See also Tables 3 and 4 for sampling site locations.

impossible to relocate. Relocation was also complicated by the fact that many of the previous sites were collected as block samples and drilled in the laboratory; hence there were no holes left behind.

[26] We took samples from 32 sites (SV01–SV32; see Table 3) during the first field season. Many of the sites from the first field season proved to have uninterpretable results owing to profound remagnetization by lightning (as described later). During a second field season in 2000 (SV50–65; see Table 4) we sought outcrops that were thought to be less likely to have been subjected to repeated lightning strikes, based on detailed surveys of the outcrop using a Bartington magnetometer and magnetic compass.

[27] Our field methods for the two field seasons were similar. Each site consisted of at least 10 field drilled samples spaced over a significant area of outcrop from a distinct cooling unit that could be demonstrated on the basis of field observations to be in situ. Each sample was oriented with magnetic (and in most cases sun) compass. In cases where sun compass measurements were impossible, each magnetic measurement was accompanied by a "backsight" check using a hand-held Brunton compass. Backsighting is accomplished by sighting from a position several meters away from the outcrop and is approximately as precise as a sun compass measurement (±2°). The cores were transported to the Scripps paleomagnetic laboratory where they were sliced yielding one to three specimens per core.

Table 3. Summary of Data From the 1999 Field Season[a]

Site(TOS90)#	Lat., N	Long., E	Age, Ma	NL	NP	k	α_{95}	D	I	Type/Pol.
sv01	35.3432	−111.7476		0	5	189	6.6	179.8	−54	III/R
sv02(3614)	35.3734	−111.7339	0.75	5	3	147	4.7	338.7	38.9	II/N
sv03	35.375	−111.731		5	5	126	4.5	343.8	52.9	III/N
sv04(3610)	35.39	−111.74		5	0	282	4.6	346.9	50.7	III/N
sv05(1221?)	35.184	−112.192		5	3	134	4.9	166.6	−51.6	III/R
sv06(1216?)	35.192	−112.193		2	3	82	9.7	351.5	−32	III/X
sv07	35.194	−112.193		7	1	320	3.1	350.3	61.8	II/N
sv08(1230)	35.17	−112.23	2.08	7	0	153	4.9	165.8	−46.1	I/R
sv09(4424)	35.442	−111.919	1.14	7	0	5.3	28.8	257.8	−74.3	II/X
sv10(2526)	35.2631	−111.827	0.44	0	5	65	13.9	188.9	−47.3	III/X
sv11(2525?)	35.263	−111.826	1.29	3	2	53	11.3	185.9	−42.5	II/X
sv12(3534A)	35.334	−111.844	0.34	1	1	26	180	78.1	60.3	III/X
sv13(2601)	35.323	−111.715	0.6	0	1	–	–	–	–	?/X
sv15(2731?)	35.249	−111.69	0.33	6	0	174	5.1	352.8	25.5	III/N
sv16(sv15?)	35.2401	−111.683		4	1	165	6.2	356	24.1	III/N
sv17	35.218	−111.443		1	1	6	180	98.2	58.3	III/X
sv18(2233A)	35.24	−112.187	4.03	5	1	985	2.2	193.1	−56.1	II/R
sv19(7833?)	35.664	−111.554	0.74	1	1	2	180	227.7	64.6	IV/X
sv20(6804?)	35.66	−111.538	1.04	1	1	7	180	7.6	30.2	IV/X
sv21(6811?)	35.645	−111.51	1.2	2	0	4	180	104.5	−2.9I	V/X
sv22(3835)	35.588	−111.517	2.43	1	1	5	180	143.9	−46	III/X
sv23(4007?)	35.482	−111.359	1.07	4	1	445	3.7	19.4	29.1	V/N
sv24(4017)	35.4673	−111.3585		8	0	128	4.9	339	63.9	III/N
sv25(6621 = 6609)	35.6243	−111.7452	1.38,1.04	2	0	2	180	221.5	24.4	III/X
sv26	35.5887	−111.6417		4	4	116	5.4	159.3	−67.8	III/R
sv27(6734)	35.5883	−111.6331	0.071	4	1	238	5.1	354	55.9	II/N
sv28(5702)	35.5838	−111.6075		2	0	3	180	344.7	74.7	IV/X
sv29	35.5526	−111.536		1	0	-	0	218.6	−23.6	IV/X
sv30(5828?)	35.5182	−111.5437		2	1	2	180	229	−52.4	III/X
sv31	35.3404	−111.5574		6	0	449	3.2	12.3	45.8	I/N
sv32(Bonita flow)	35.3708	−111.5393	1080AD	5	0	298	4.4	344	60.3	I/N

[a] Site is our site numbers. TOS90 is the name given by *Tanaka et al.* [1990] for the same unit. Ages are from *Tanaka et al.* [1990]. NL are the number of directed lines and NP are the number of best-fit planes included in the average. k, α_{95} are the estimated Fisher parameters (see text). D, I are the declination and inclination, respectively. Type is a classification based on the behavior during demagnetization (see text), and Pol. is the polarity. X are sites not meeting minimum criteria (see text).

Table 4. Summary of Data for the 2000 Field Season[a]

Site(TOS90#)	Lat., N	Long., E	Age, Ma	NL	NP	k	α_{95}	D	I	Type/Pol.
sv50	35.3508	−112.4074		2	1	1	180	266.8	−40.9	V/X
sv51(2Z10)	35.3108	−112.374		5	0	3	56.5	199.9	−36.9	IV/X
sv52	35.2211	−112.2719		5	0	223	5.1	181.6	−55.6	III/R
sv53(1102)	35.2221	−112.2539		4	0	4	49.7	177.5	−32.6	V/X
sv54(2110A)	35.3002	−112.2707		5	0	30	14.1	175.3	−46.7	III/X
sv55	35.3195	−112.1798		5	0	314	4.3	181.2	−46.3	I/R
sv56	35.3417	−112.1576		5	1	5	31.9	169.4	−42.3	III/X
sv57	35.26	−112.2436		6	0	23	14.1	197.7	−47	III/X
sv58	35.2603	−112.2436		5	0	594	3.1	181.4	−61.7	I/R
sv59	35.2542	−112.0066		5	0	636	3	177.1	−51.1	III/R
sv60	35.2535	−111.9136		3	0	17	31.2	195	−28.3	III/X
sv61	35.2031	−111.7638		4	1	8	29.6	1.6	70.5	II/X
sv62	35.2078	−111.7706		1	3	21	28.1	311.9	44	III/N
sv63	35.2582	−112.0835		5	0	602	3.1	351.6	44.8	I/N
sv64	35.2569	−111.9315		7	0	137	5.2	169.1	−52.9	III/R
sv65	35.4088	−111.4146		4	2	93	7.3	29.6	55	III/X

[a] See footnote for Table 3.

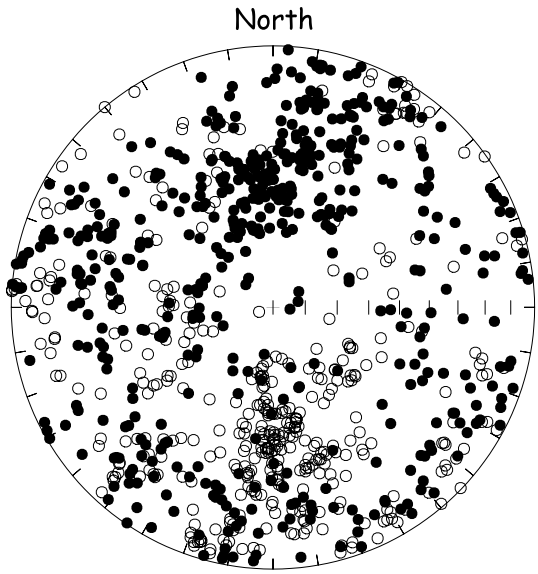


Figure 5. Equal area projection of NRM directions from all specimens. Solid (open) symbols are lower (upper) hemisphere projections.

[28] The natural remanent magnetizations (NRMs) of the SV specimens were measured using a 2G Enterprises cryogenic magnetometer, or a Molspin "Minispin" magnetometer if the NRM exceeded about 0.5 A/m. The NRM directions from all our specimens of the San Francisco volcanics are plotted in Figure 5. These are essentially randomly oriented.

[29] At least five specimens from at least five separately oriented samples were subjected to AF demagnetization from each site. Steps were as follows: 5, 10, 20, 30, 40, 50, 60, 70, 80, 100, 120, 140, 160, and 180 mT. At peak fields higher than about 80 mT, the specimens underwent so-called "double demagnetization" whereby after measurement at a given step, the AF treatment was repeated along the opposite axes and remeasured. In these cases, we use the vector mean of the two measurements. Double demagnetization effectively eliminates the (rare) problem of ARM acquisition during AF demagnetization. In addition, at least one specimen from each site was subjected to step-wise thermal demagnetization in 50° steps starting at 100°C to 500°, followed by 25° steps to the maximum unblocking temperature. All measurement data are available in the file sv.xls.

[30] There was a tremendous variety in behavior of the different sites, reminiscent of the troubles described in the previous studies. We classified our sites into five categories based on their NRM directions and behavior during AF and thermal demagnetization.

3.1. Type I

[31] Figure 6 illustrates a site typical of our "Type I" sites. There are a total of six of these in our entire collection of 47. The NRM directions are well grouped and AF and thermal demagnetization both yield straightforward univectorial demagnetization behavior. We calculate best-fit lines using principal component analysis [*Kirschvink*, 1980] and define an acceptable line as having at least four consecutive demagnetization steps and a maximum angle of deviation (MAD) of $\leq 5°$ (rounded to the nearest integer).

3.2. Type II

[32] Figure 7 shows a typical example of our "Type II" sites. There are seven of these in our collection. The NRM directions are spread along a great circle (Figure 7d). Thermal demagnetization for Type II behavior yields straight-line decay to the origin in an apparently meaningless direction (Figure 7a). AF demagnetization, however, does a good job of erasing a soft component. Some specimens reach a stable end-point and the data can be interpreted as a best-fit line. Other specimens never reach a stable end-point. For these, we calculate a best-fit great circle (Figure 7c) using principal component analysis with at least four consecutive demagnetization steps. Acceptable planes must have MADs $\leq 5°$ (rounded to the nearest integer). The site mean (triangle in Figure 7e) is based on a combination of best-fit lines (dots) and planes (great circles) using the technique of *McFadden and McElhinny* [1988].

[33] We suspected that the large degree of scatter manifested by the data in Figure 7 was caused by lightning (see, e.g., *Sakai et al.* [1998] for more discussion). For that reason, we located each sample along the outcrop and measured its distance to a nearby tree that had obviously been hit by lightning (see Figure 8). In the laboratory, we induced an

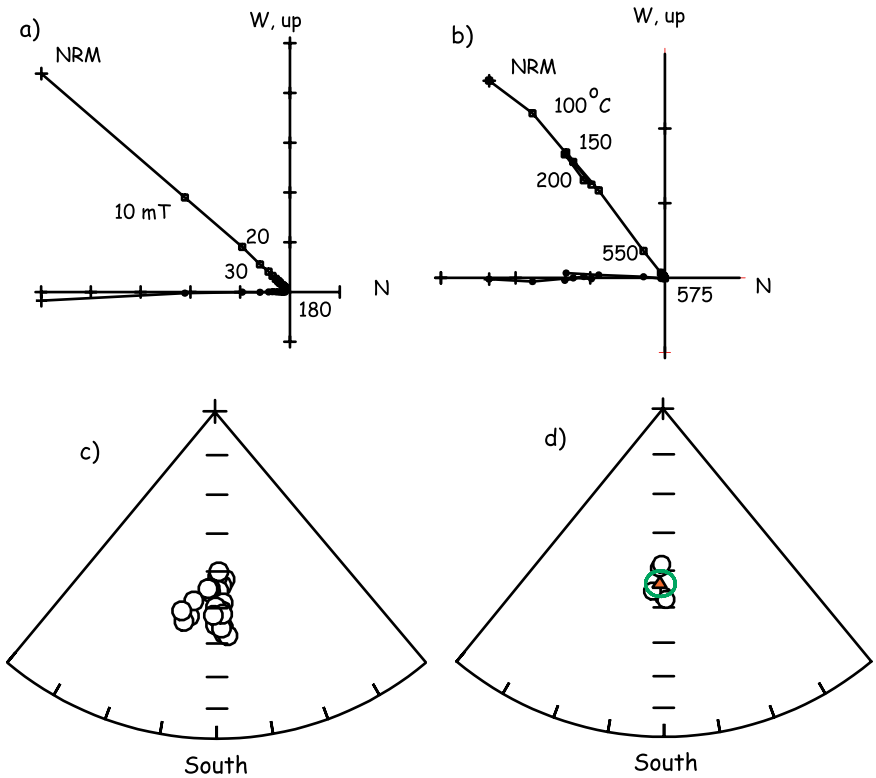


Figure 6. Typical "Type I" site (SV55). a) AF demagnetization of specimen SV55h1 shows straightforward, univectorial decay to the origin. b) Thermal demagnetization of specimen SV55b1. Behavior during thermal demagnetization is similar to during AF demagnetization. c) All NRM specimen directions from the site. These are already well grouped before demagnetization. d) Directions of best-fit lines defining characteristic directions (circles) for data similar to those shown in a) and b), site mean (triangle), and α_{95} (circle).

isothermal remanent magnetization (IRM) using an ASC impulse magnetizer, sufficient to recreate the observed NRM, inferring that this was proportional to the original inducing field although the original IRM is likely to have relaxed somewhat since it was acquired; hence the laboratory impulse field is a minimum estimate. The outcrop pattern of compass deflection (backsight declination minus the outcrop declination for the sample orientation), NRM intensity and inferred minimum inducing fields are summarized in Table 5. The magnetizations of the samples nearest the tree are an order of magnitude larger than those farther away. The high magnetization of the outcrop nearest the tree results in a large deflection of the magnetic compass near those sampling locations. Also, the NRM directions of specimens from the same sample in samples nearest the tree are different by more than a few degrees, which is not the case in samples far from the tree. Finally, the high NRMs in samples near the tree are very "soft" with median destructive fields typically less than 10mT.

[34] We plot the inferred minimum inducing field B versus distance from the tree in Figure 9. If we treat the lightning strike as a current in an infinite wire, the magnetic field B produced is given as a function of distance (r) by

$$B = \frac{\mu_o I}{2\pi r}$$

where I is current and μ_o is the permeability of free space. The best-fit current to the data shown in Figure 9 is approximately 300,000 Amps. This is in the range of the strongest lightning strikes [*Krider and Roble*, 1986].

[35] (In the case shown in Figure 8, we believe that we have identified the location of the lightning strike responsible for the observed remagnetization. However, in most cases, such an identification will be

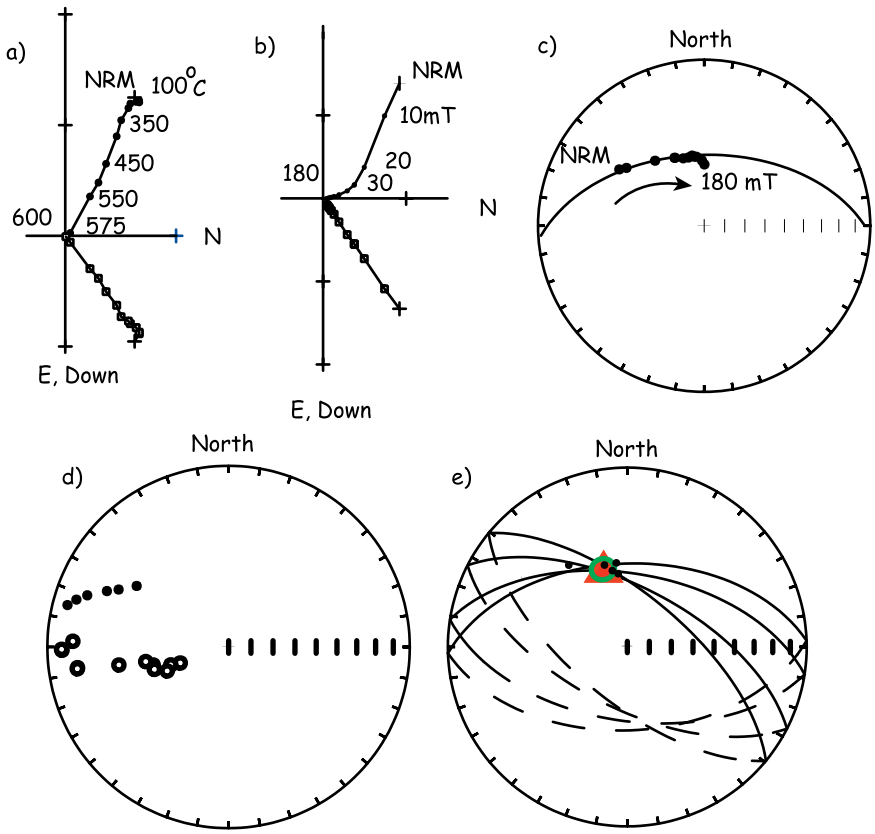


Figure 7. Typical "Type II" site (SV03). a) Thermal demagnetization of SV03b2. b) AF demagnetization of SV03a1. c) Same data as in b) but in equal area projection. The demagnetization data travel along a great circle path toward a direction common to all AF demagnetized specimens from the site. d) All NRM specimen directions from the site. e) Directions of best-fit lines (circles), best-fit great circles (lines), site mean (triangle). Solid (dashed) lines are lower (upper) hemisphere projections.

impossible because the surface evidence for the strike (in this case, a tree) does not last for many years (or may never have left a visible trace at all), while the remagnetization may last a very long time.)

3.3. Type III

[36] The site shown in Figure 10 is typical of our "Type III" sites of which we found 24 in our collection. The NRM directions are scattered. AF is most effective in isolating a principal component. Best-fit lines and planes result in much reduced scatter and we were able to calculate acceptable site means for some of the Type III sites. This behavior is likely to also be the result of multiple lightning strikes.

3.4. Type IV

[37] Figure 11 shows a typical "Type IV" site of which there are five. Type IV sites are characterized by highly scattered NRM directions. Unlike Type II and III sites, demagnetizations by both AF and thermal treatments result in univectorial decay to the origin. The principal components remain scattered and site means have large α_{95}s and low ks.

3.5. Type V

[38] "Type V" sites (see Figure 12) are characterized by scattered initial NRMs like the Type III and IV sites. There are five of these sites in our collection. Specimens from Type V sites differ from the others in that AF demagnetization shows complicated multicomponent behavior with several great circle tracks. Thermal demagnetization is usually univectorial. We interpret this behavior as the result of several successive lightning strikes near the samples.

3.6. Summary of New Data

[39] We have made all the demagnetization data available as well as best-fit lines, planes and site

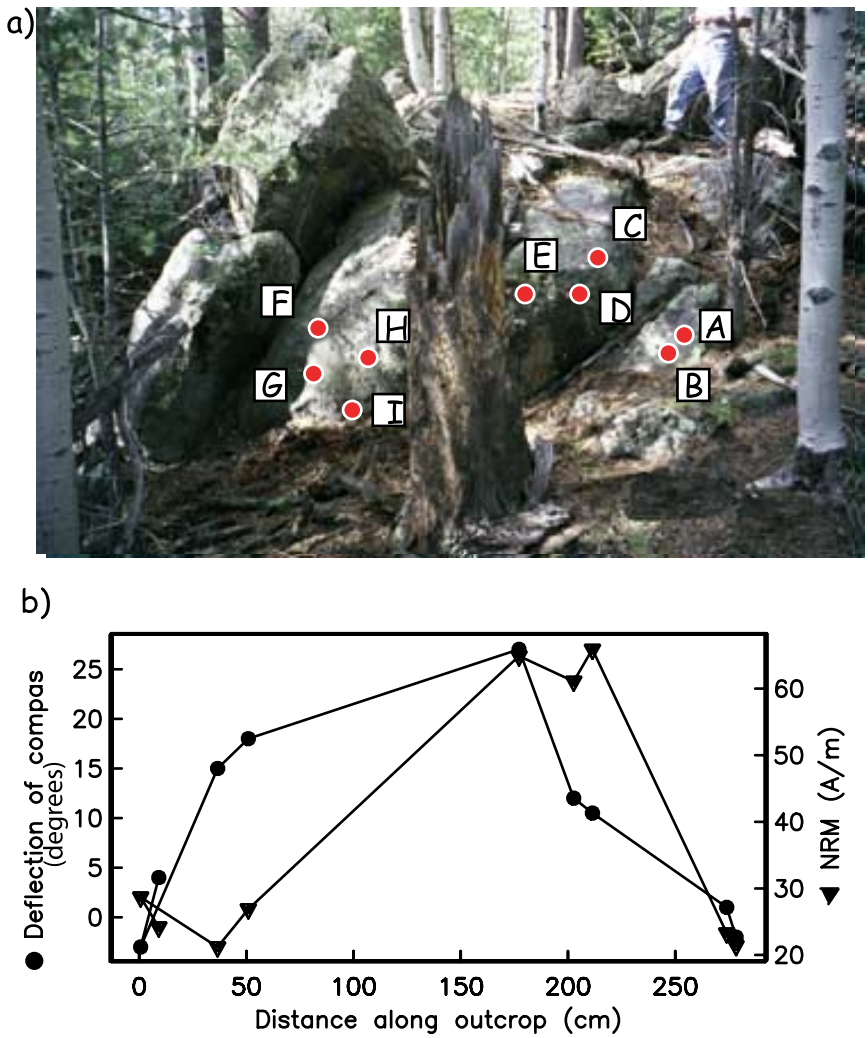


Figure 8. a) Outcrop photograph of Site SV03 with location of samples labeled (see Table 5). b) Deflection of magnetic compass at outcrop and NRM of samples (see Table 5).

means in the file sv.xls. Site statistics are summarized in Tables 3 and 4. In general AF demagnetization provided the most reliable results, but the overall yield of high quality data was disappointingly small. Of the 47 new sites only 28 yielded mean directions with Fisherian α_{95} of less than 25°, a rather loose quality criterion. Using an alternative measure, we found 22 sites with $k \geq 100$, of which 12 are normal and 10 reverse. We plot the site mean directions for all sites with at least two separately oriented samples in Figure 13. Those with estimates of $k \geq 100$ are plotted as squares while those from the less well grouped sites are plotted as circles. In general the better determined sites are less scattered and are more

Table 5. Compass Deflection, NRM Intensity, and Inferred Remagnetizing Field for SV03 Specimens[a]

Sample	X, m	R, m	Δ Dec.	NRM, A/m	*B*, mT
A	2.78	1.87	−2	21	
B	2.74	1.82	1	23	13.5
C	2.11	1.76	10.5	66	
D	2.0	1.56	12	61	21
E	1.77	1.54	27	65	23.9
F	0.0	2.52	−3	29	
G	0.09	2.50	4	24	9.2
H	0.51	2.16	18	27	10
I	0.368	2.28	15	21	8.6
J	0.6	2.02	-	22	9

[a] X is distance along outcrop, R is distance from tree shown in Figure 8, Δ Dec. is the deflection of the magnetic compass determined from comparison of magnetic declination near the outcrop and the backsighted declination, *B* is the inferred remagnetizing field from the lightning strike, based on laboratory-induced IRM.

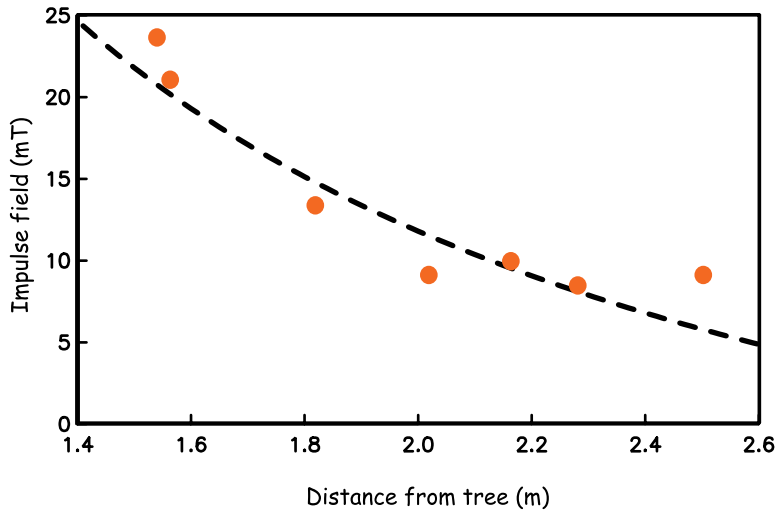


Figure 9. Impulse field required to reproduce the NRM intensity as a function of distance from the tree shown in Figure 8. Dashed line is best-fit to the data assuming that the tree shown in Figure 8 was the site of a remagnetizing line current (lightning bolt) of 300,000 Amps.

likely to be consistent with the direction expected from a GAD field.

4. Discussion

[40] We turn now to an analysis of the combined new and published data sets. As we noted, the published data are quite scattered and the question arises as to whether the dispersion is geomagnetic in origin or results from other sources. Potential contaminating signals include: unaccounted-for rotation from synvolcanic or postvolcanic emplacement after cooling through the Curie temperature, reheating by subsequent volcanic eruptions, lightning, viscous remanence, weathering or chemical alteration. In the following discussion we evaluate various possible sources and their likely contributions to the existing data set.

4.1. Tectonic Disruption

[41] Directions from areas unaffected by tectonics are shown in Figure 14a where data from all the previously published sites are combined with the new data set. Because the region of interest is rather large, the expected GAD direction at the various sampling sites differs by several degrees (see Table 2). In such cases it is appropriate to consider the data relative to the expected GAD direction at the sampling site. For this purpose, it is useful to use the transformation proposed by *Hoffman* [1984], whereby each direction is rotated such that the direction expected from a geocentric axial dipole field (GAD) at the sampling site is the center of the equal area projection. This is accomplished as follows:

[42] Each direction is converted to Cartesian coordinates (x_i) by

$$x_1 = \cos D \cos I$$

$$x_2 = \sin D \cos I$$

$$x_3 = \sin I$$

[43] These are rotated to the new coordinate system (x_i') by

$$x_1' = \left(x_1^2 + x_3^2\right)^{1/2} \sin(I_d - \theta)$$

$$x_2' = x_2$$

$$x_3' = \left(x_1^2 + x_3^2\right)^{1/2} \cos(I_d - \theta)$$

where I_d = the inclination expected from a GAD field ($\tan I_d = 2\tan \lambda$), λ is the site latitude, and θ is the inclination of the paleofield vector projected onto the N-S plane ($\theta = \tan^{-1}(x_3/x_1)$). The x'_i are then converted to D', I' by the usual transformations. Using the D', I' transformation avoids the distortion of the population of directions inherent in conversion to virtual geomagnetic poles

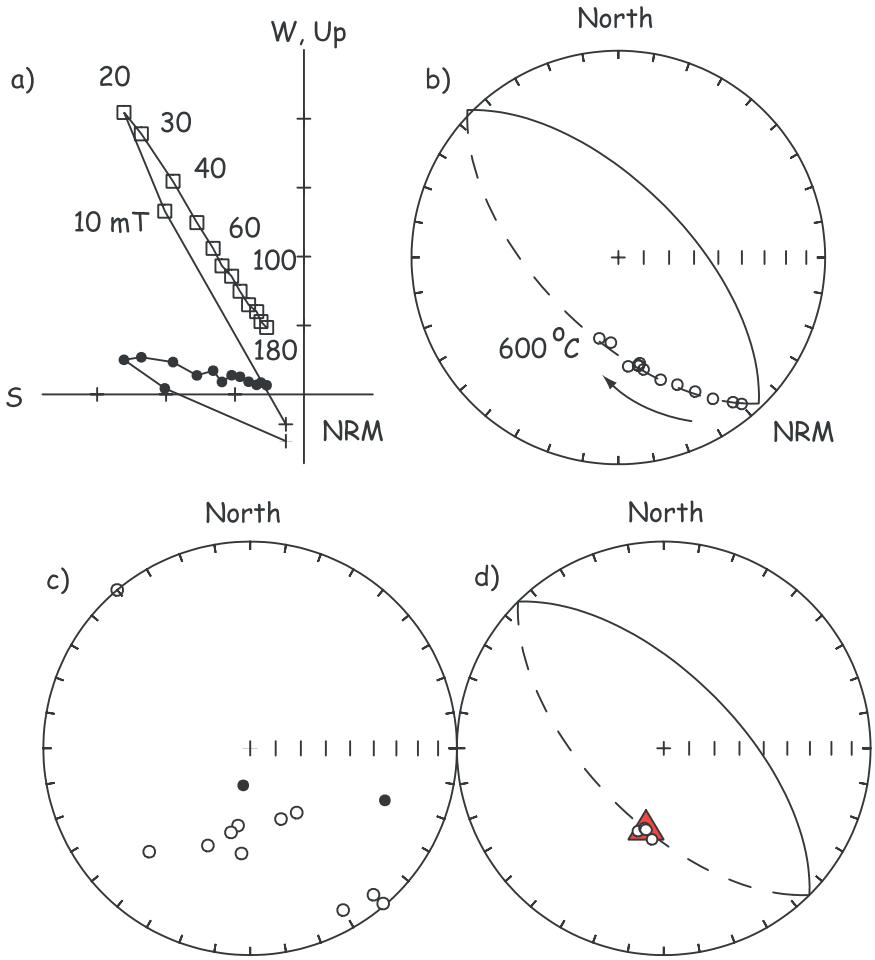


Figure 10. Typical "Type III" site (SV18). a) AF demagnetization of SV18f1. The characteristic direction is isolated by about 20 mT. b) Thermal demagnetization of SV18i2. The data travel along a great circle path toward a direction common to all specimens (see d). c) All NRM specimen directions from the site. These are quite scattered, with no apparent pattern. d) Directions of best-fit lines (circles), best-fit great circles (lines), site mean (triangle) and α_{95} (circle). Solid (dashed) lines are lower (upper) hemisphere projections.

(VGPs), while eliminating the dependence of inclination on latitude. We can now compare directly the directions of Figure 2 for tectonically suspect regions with those from unaffected areas in Figure 3 as well as to the expected GAD directions (see Table 2). The two data sets are quite different: the tectonically suspect regions have an inclination anomaly in both the normal and reverse data of $\sim 10^\circ$, while when only those from stable areas are considered the overall average inclination lies within 2° of the expected direction. Because large average deviations from GAD are correlated with suspected tectonic activity we eliminate the 164 sites from studies #1, #2, and #8 from further discussion and consider only the remaining 701 directions. These are plotted in Figure 14 both as D, I (Figure 14a) and as D', I' (Figures 14b and 14c).

4.2. Role of Sub-Curie Temperature, Synemplacement Rotation

[44] Volcanic emplacement is often a violent act. Aa flows may carry relatively cool blocks some distance on their surfaces. Many lava flows crack upon cooling. Volcanic vents undergo inflation and deflation. These and many other factors can result in the possibility of rotation of the rock unit after cooling through the Curie temperature. Many of these problems can be detected and avoided through careful field work, but the paleomagnetic

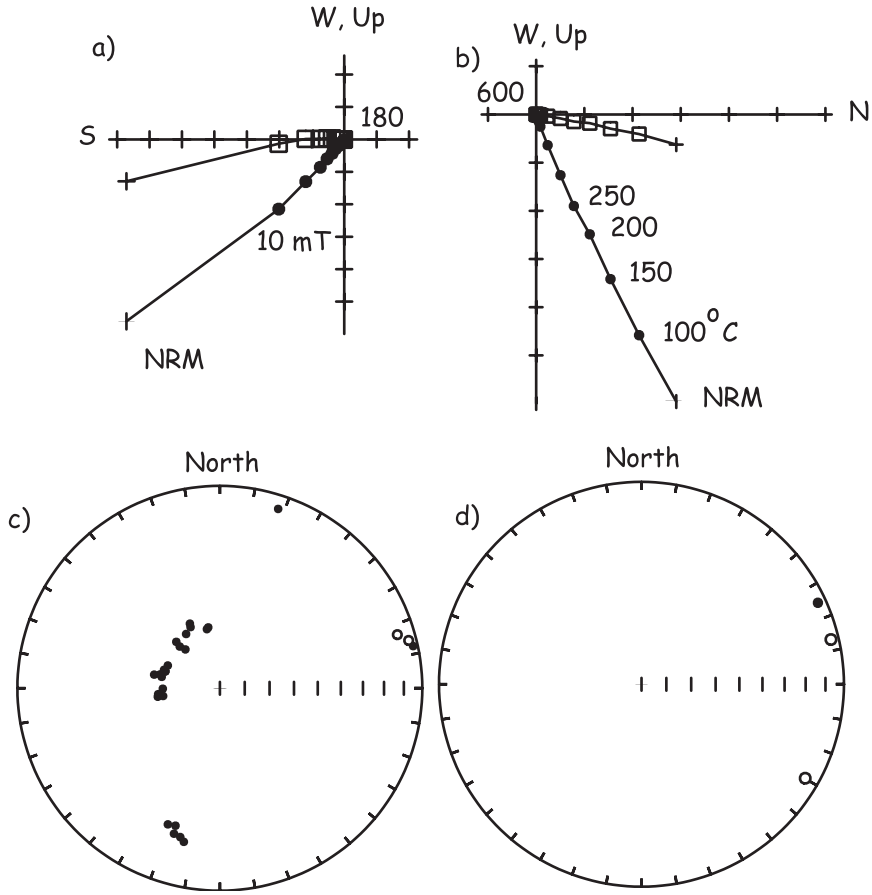


Figure 11. Typical "Type IV" site (SV21). a) AF demagnetization of SV21i1. No significant directional change is observed up to 180 mT. b) Thermal demagnetization of SV21g2. As for AF, no significant change in direction is observed up to the maximum blocking temperature between 550 and 600°C. c) All NRM specimen directions from the site. These are quite scattered, with no apparent pattern. d) Directions of best-fit lines (circles). No significant mean direction could be calculated.

data set reported here now comprises studies done by numerous investigators and field assistants over a span of nearly four decades. Volcanic vents, domes and pyroclastic breccias are more likely to undergo undetected post cooling rotation than are dikes, tuffs, and lava flows. We have assigned all but a few of the 701 data points an emplacement style based on the information provided in the publications. We separate these into two groups. Group "V" comprises vents (382), domes (54) and breccias (2) and group "F" comprises flows (251), dikes (6), and tuffs (3). We found no statistically significant difference between these two groups. They have comparable Fisher dispersion parameters of about 15 and statistically indistinguishable mean directions.

[45] As there are many reasons for any of these emplacement mechanisms to give rise to scattered directions, it may be more appropriate to exclude data on the basis of within site reproducibility, rather than emplacement mode. We examine the effect of within site reproducibility in the following.

4.3. Within Site Reproducibility

[46] There are several commonly used proxies for data quality that reflect the accuracy and reproducibility of the field orientation methods and laboratory measurements that are combined to produce the site mean direction. These proxies include the following parameters: N, the total number of separately oriented samples measured from the site, α_{95}, the *Fisher* [1953] cone of 95% confidence in

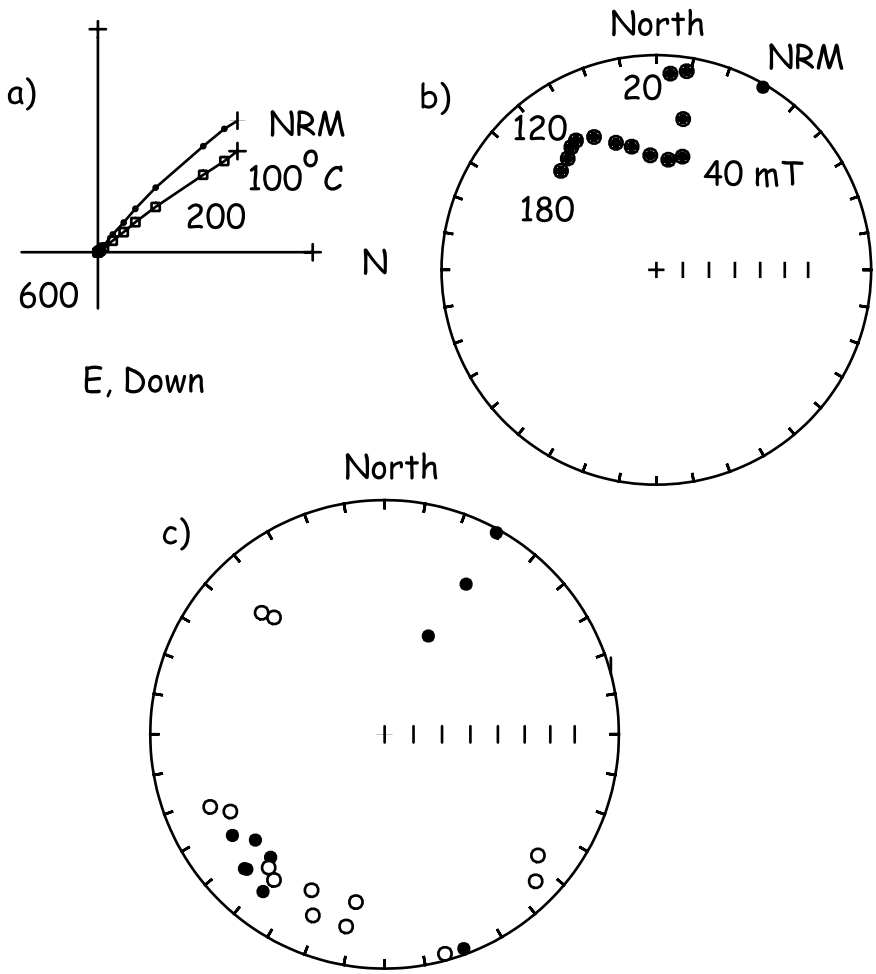


Figure 12. Typical "Type V" site (SV29). a) Thermal demagnetization of SV29d2 results in univectorial decay to the origin. b) AF demagnetization of SV21g2. This specimen has at least three components evident in the demagnetization data. c) All NRM specimen directions from the site. These are quite scattered, with no apparent pattern.

the mean, and the *Fisher* [1953] precision parameter (κ). The latter is estimated by $k = \frac{N-1}{N-R}$ where R is the resultant vector assuming all the directions to be unit vectors. α_{95} is estimated by

$$\alpha_{95} = \cos^{-1}\left[1 - \frac{N-R}{R}\left[\left(\frac{1}{0.05}\right)^{\frac{1}{(N-1)}} - 1\right]\right]$$

or, as suggested in many textbooks [e.g., *Irving*, 1964; *McElhinny*, 1973; *Butler*, 1992],

$$\alpha'_{95} \approx \frac{140^\circ}{\sqrt{kN}}.$$

The latter was shown to be a poor approximation for α_{95} for data sets with $k \leq \sim 25$ by *Tauxe et al.* [1991], and *McElhinny and McFadden* [2000] proposed the use of another approximation which may be less biased:

$$\alpha''_{95} = 140^\circ/\sqrt{kR}.$$

Authors typically do not specify which approximation has been used to estimate α_{95}, a practice that could lead to a great deal of confusion.

[47] Because α_{95} is a measure of confidence in the calculated mean direction it is strongly dependent on N. k is intended to reflect the variance of the underlying distribution of directions which should be nearly independent of N (although in practice its accurate estimation may require a minimum

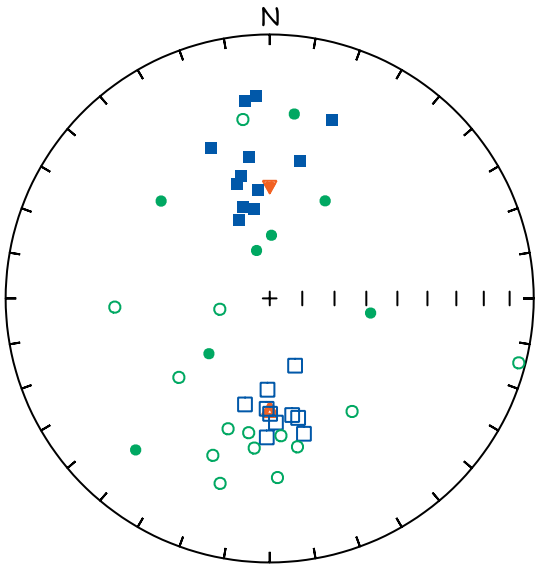


Figure 13. a) Equal area projections of site means with $N \geq 2$. Solid (open) symbols are lower (upper) hemisphere projections. Squares are sites with $k \geq 100$ and circles are those with $k < 100$.

number of samples as discussed later). In secular variation studies it is common to correct for the effects of within site scatter in an attempt to recover the purely geomagnetic contribution [see, e.g., *McElhinny and Merrill*, 1975]. However, the aforementioned proxies of this scatter can be biased. It is possible to achieve a small α_{95} by using a large number of observations, while the underlying distribution is highly scattered. It is also possible to achieve a large value of k with an erroneous direction by, for example, sampling only one block from the top of a severely rotated aa flow. Hence, measuring a large number of samples does not guarantee a meaningful result if all the samples come from the same restricted area. We investigate the potential for bias and its consequence by eliminating data using successively more rigorous criteria based on k and α_{95} individually.

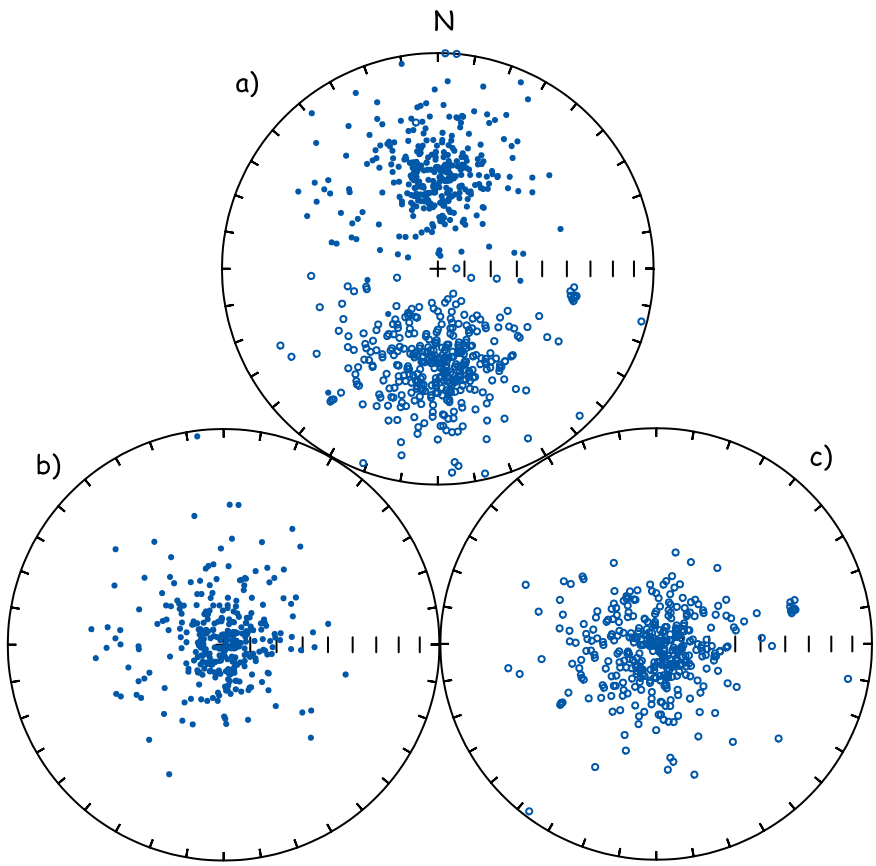


Figure 14. Equal area projections of all data unaffected by tectonic disturbance from the southwestern United States. a) Symbols same as in Figure 2. b) Normal data from a) projected on to the expected directions from a Geocentric Axial Dipole field at each sampling location (D', I' projection of *Hoffman* [1984]). c) same as b) but for reverse directions.

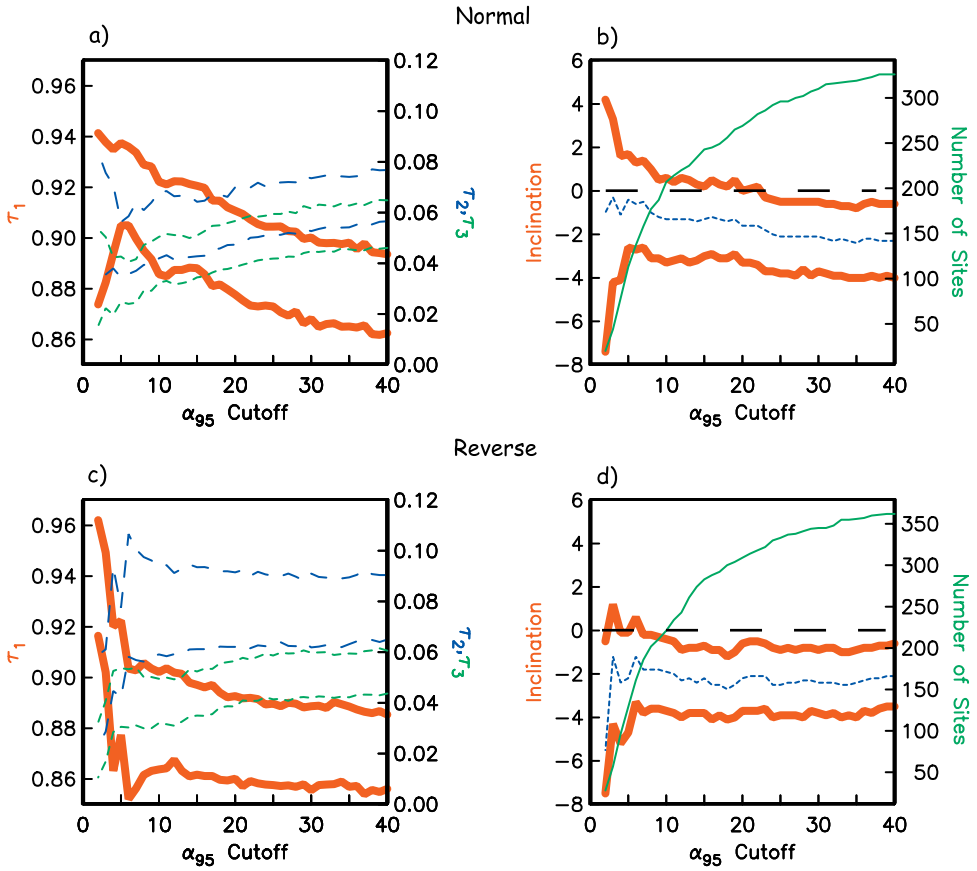


Figure 15. The effect of using α_{95} as a rejection criterion. a) 95% confidence bounds for the eigenvalues in decreasing order τ_1 (heavy lines), τ_2 (dashed lines), and τ_3 (dotted lines) for the normal data shown in Figure 14b. b) Confidence bounds for the normal data on the inclination anomaly plotted so that negative values correspond to directions more shallow than expected from GAD. Dashed line is the average inclination anomaly. Thin solid line is the number of sites remaining at each cutoff value. c) Same as a) but for the reverse data of Figure 14c. d) Same as b) but for reverse data.

[48] We begin by examining the effect of using α_{95} as a selection parameter. To assess its influence we take the D', I' data from Figures 14b and 14c, which is a compilation of all published and new data from the southwestern United States excluding the three data sets from areas suspected of tectonic activity. Taking the normal and reverse modes separately, we eliminate data from sites with successively smaller values of α_{95}. We evaluate the principal components of the resulting data set (in the manner of, for example, *Kirschvink* [1980]). The eigenparameters of interest here are the eigenvalues from largest to smallest (τ_1, τ_2, and τ_3) and the direction of the principal eigenvector. The 95% confidence bounds on the eigenparameters are estimated using 1000 bootstrapped pseudosamples of each selected data set (see *Tauxe* [1998] for a complete discussion of bootstrapping in paleomagnetism). The less scattered the data set, the larger τ_1. If the data distribution is axially symmetric about the principal axis, then the confidence bounds on τ_2 and τ_3 will overlap. We have rotated the principal direction such that a negative value for inclination means the average is more shallow than the expected direction from a GAD field for either normal or reverse directions.

[49] We plot the results of the principal component analysis in Figure 15. In Figure 15a we plot the confidence bounds for the eigenvalues as a function of increasing α_{95} cutoff value. Not surprisingly, the scatter is greatly reduced (τ_1 increases, while τ_2 and τ_3 decrease) as the cutoff value of α_{95} decreases because this excludes the less well

determined site means. The distribution of the normal data cannot be distinguished from one with axial symmetry regardless of imposed cutoff value (the confidence bounds on τ_2 and τ_3 overlap). In Figure 15b, we plot the resulting inclination anomaly. For the entire normal data set, there is a small negative anomaly (less than 2 on average), whose statistical significance disappears on excluding sites with α_{95} of greater than about 20°. Also shown is the number of sites remaining (N_{sites}) as a function of α_{95}. The existence of a small negative anomaly is consistent with the predictions of *Creer* [1983], who pointed out that use of unit vectors will introduce a shallow bias in average directions.

[50] The reverse data are somewhat different from the normal data, as shown in Figures 15c and 15d. The entire reverse data set cannot be considered axially symmetric, except by excluding all sites whose α_{95}s are greater than about 5°, at which point the inclination anomaly is once again not significantly different from zero. The average inclination anomaly is slightly larger for the reverse polarity data than for normal. In Figures 15b and 15d we see that most of the data would be rejected by using a cutoff of 5°, as used by, for example, *Johnson et al.* [1998] and *Tauxe et al.* [2000].

[51] We performed a similar analysis on the data using k as the selection criterion. However, in this case, we also investigate the behavior of the data set in which normal and reverse directions are combined as well. The results are shown in Figure 16 and are broadly similar to the results using α_{95} when the relationship between k and α_{95} is taken into account. The cutoff value that produces an axially symmetric data set is approximately 100. Small but systematic inclination anomalies appear to persist, although they are once again barely statistically significant. Because α_{95} can be manipulated by using large values of N, we prefer to use $k \geq 100$ as a selection criterion. We should note here that we have no a priori prejudice against asymmetry in the data distributions, but are deeply skeptical of any asymmetry that is only manifest when apparently lower quality data are included in the analysis.

[52] Previous studies of paleosecular variation have estimated the scatter in the equivalent VGPs rather than using principal component analysis on D', I'. as done here. Scatter is quantified with the parameter S_p (e.g., MM97) which is given by

$$S_p^2 = \sum_i^N \frac{\delta_i^2}{N-1}$$

where S_p is the total angular dispersion, δ_i is the angle between the ith VGP latitude and the spin axis. Both between site dispersion (S_f) and within-site dispersion (S_w) normalized by the average number of samples per site ($\bar{N}$) contribute to the total dispersion as given by:

$$S_f^2 = S_p^2 - \left(S_w^2/\bar{N}\right)$$

S_w is in fact the circular standard deviation (CSD) of the within-site VGPs, defined by *Irving* [1964] as

$$CSD \simeq \frac{81^\circ}{\sqrt{k}}. \quad (1)$$

where k here is the average estimated κ for the VGPs. We prefer the principal component approach because it avoids the distortions associated with the VGP transformation and allows the ellipticity of the data distribution to be examined as well as the overall scatter. However, to facilitate comparison with earlier studies, we show the same analysis as in Figure 16, but calculating S_p in Figure 17. As expected, the behavior of S_p is the inverse of that for τ_1. We note that in the normal polarity data there is a systematic increase in scatter S_p as k increases above 130, which is also seen in the wider confidence bound for τ_1 for the principal component analysis. The reason for this is not well understood by us, but may reflect bias from the diminishing size of the data set with increasingly rigorous cutoff. However, no such effect is seen in the reverse data set, for which the dispersion steadily decreases with increasing k.

[53] The question now arises as to what the effect of the number of samples has on our ability to estimate k. First, if there are too few samples, or the samples are not distributed over a wide outcrop extent, k will be a poor estimate of the true scatter

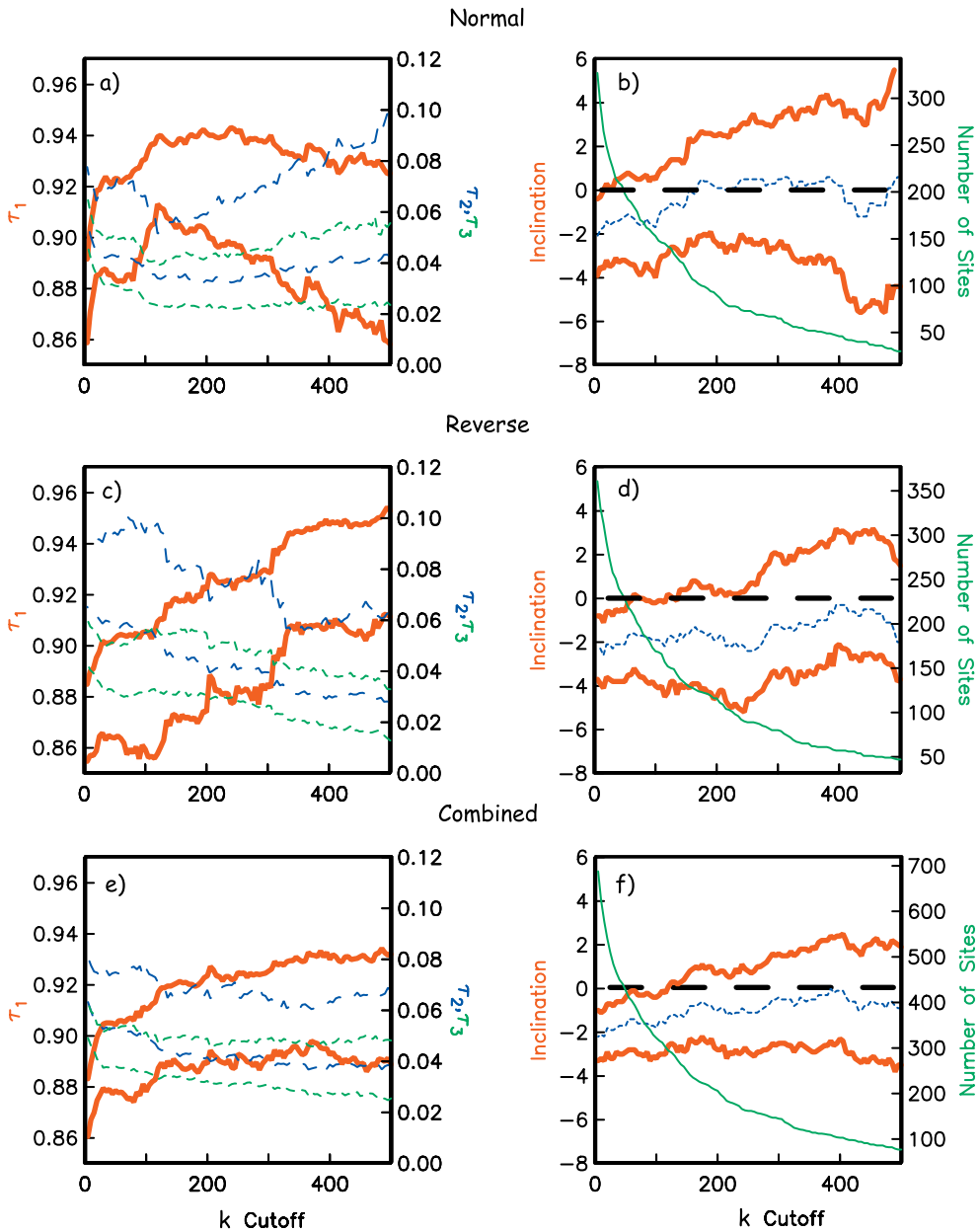


Figure 16. a–d) Same as Figure 15 but using k as a cutoff value. e) and f) same as a) and b) except using combined normal and reversed data.

in the rock unit. Neglecting the effects of orientation error and outcrop inhomogeneity, we examine the issue of the number N of samples required to estimate k reliably. The question is, "At what value of N is the resulting estimated k independent of N?" Because k typically ranges over many decades, it is advantageous to convert it to the circular standard deviation (*CSD*) using equation 1. A *CSD* of 8.1 is therefore equivalent to $k = 100$.

[54] To investigate the dependence of *CSD* on N, we draw 100 random data sets from a *Fisher* [1953] distribution using the **fisher** program described by *Tauxe* [1998] and available online at ftp://sorcerer.ucsd.edu/pub/pmag. The *CSD* is then calculated for each synthetic data set and plotted in Figure 18. The 95% confidence bounds are shown as the solid lines. Based on this analysis, it seems that the best selection criteria for the data set

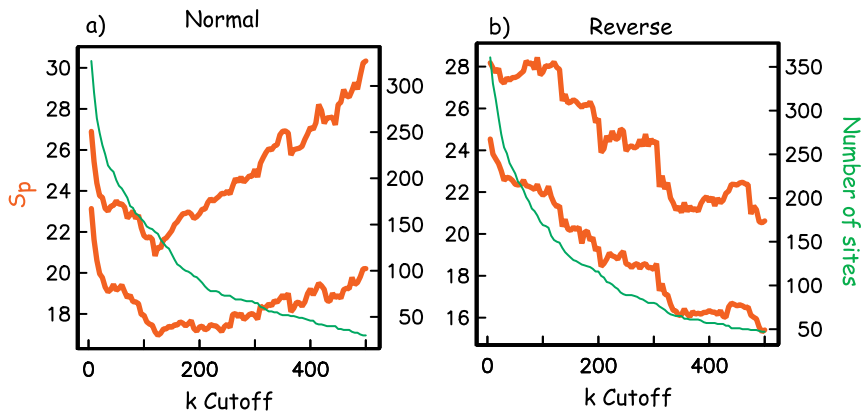


Figure 17. Same as in Figure 15a but calculating S_p on the VGPs instead of principal component analysis on the directions.

compiled here is to use a cutoff value of $k = 100$ and $N = 5$.

4.4. How Many Sites Are Enough to Estimate TAF and PSV?

[55] In Figures 15b and 15d we see that over two thirds of the data would be rejected by using an α_{95} cutoff of 5°, a number advocated in some recent TAF papers [e.g., *Johnson et al.*, 1998; *Tauxe et al.*, 2000]. Similarly, the selection criterion of $k \geq 100$ rejects over half the data. Increasingly stringent selection criteria are liable to eliminate large fractions of a data compilation containing information collected by techniques that have been steadily improving over the past 40 years. What number of sites is required to achieve a stable estimate of the TAF and paleosecular variation? This issue is different from those already discussed in that it is no longer a question of data quality, but one of how much variation occurs in the geomagnetic field, and how many temporally distinct observations might be required to characterize the variability. Earlier proposed selection criteria (e.g., MM97) have suggested that, while more are obviously desirable, as few as five sites (nine in the case of dual polarity) might be adequate to obtain an estimate of scatter. The adequacy of this estimate for both TAF and PSV must depend on how well the range of magnetic field variability has been sampled.

[56] We investigate the issue of the minimum number of sites (here designated N_{site}, not to be confused with the previous section's N, the number of samples per site) using Monte Carlo type simulations from CJ98 and CJ98.nz, two representative statistical models for paleosecular variation (PSV) [*Constable and Johnson*, 1999]. These models can be used to predict the statistical distributions of any paleomagnetic observable, such as the average field direction, or dispersion in VGP position. A limitation of using models is that they may not describe the geomagnetic field

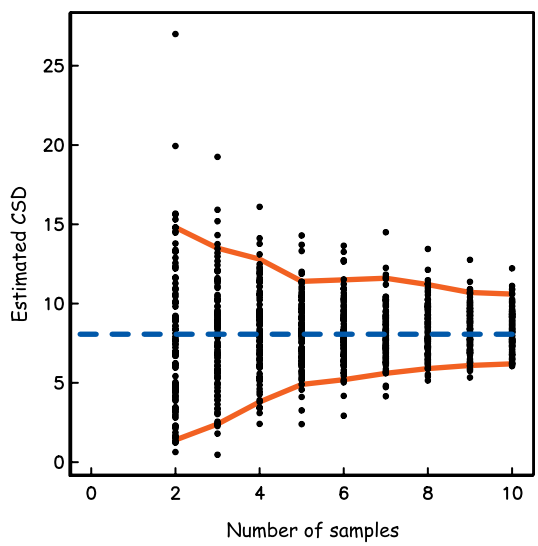


Figure 18. Estimated *CSD* for 1000 sites with N samples, drawn from a Fisher distribution with $\kappa = 100$. The equivalent *CSD* of 8.1 is shown by the dashed line. The estimated *CSD* becomes relatively independent of the number of samples for $N \sim 5$. The heavy lines are the 95% confidence bounds.

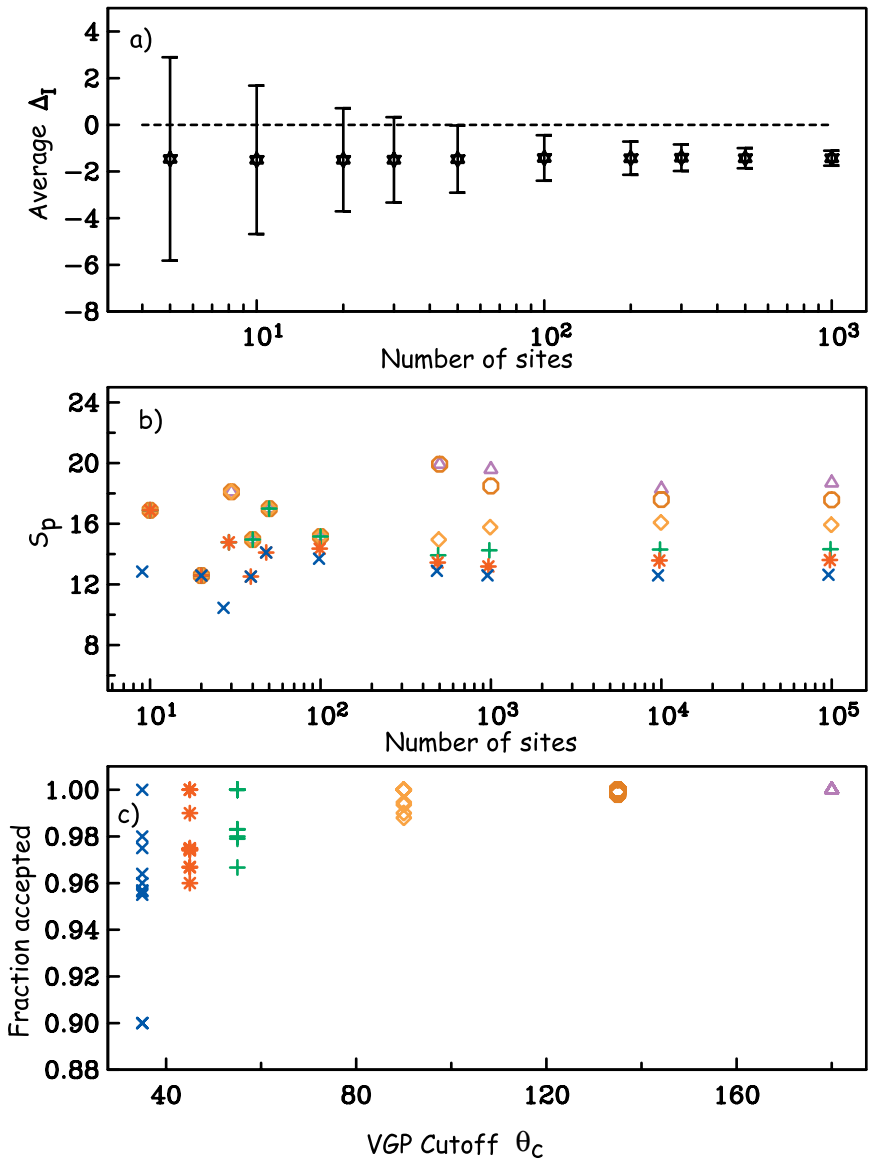


Figure 19. Results from 1000 simulations of CJ98.GAD PSV. a) Average of 1000 inclination anomalies, $\Delta\bar{I}$ (stars) from the vector average of the directional data, error bars represent one standard error for the individual estimates based upon N_{site}. b) Dispersion about the rotation axis, S_p as a function of number of sites; symbols correspond to the values of Θ_c shown in (c). $\Theta_c = 45°$ was used in (a). (c) Influence of Θ_c on the fraction of data accepted for CJ98.GAD.

adequately (indeed if they did there would hardly be a need for the current study), but these models are at least based on existing data, and can serve as a guide in our efforts. For the purposes of simulation we neglect the influence of rock magnetic and orientation errors. Because the size of any non-axial-dipole contribution to the time averaged geomagnetic field remains controversial we simplify the models so that all TAF contributions are zero except for g_1^0, in other words the vector average field is GAD. We call these variants of the published model CJ98.GAD and CJ98.nz.GAD for zonal and nonzonal models, respectively. Figure 19 provides a summary of the influence of the number of data on a) average inclination anomaly ($\Delta\bar{I}$) derived from unit vectors, b) estimated S_p and c) the influence of the excluding points deemed transitional or excur-

sional on the fraction of observations that get accepted.

[57] We plot results from 1000 simulations of CJ98.GAD in Figure 19a, with the number of sites ranging from 5 to 1000. All data with $\Theta_c > 45°$ were eliminated from the average, a procedure commonly implemented in paleomagnetic studies that are not concerned with transitional fields (more on this later). The scalar mean of 1000 simulations of $\Delta\bar{I}$ calculated for the vector average of N_{site} directional data is quite consistently estimated as between -1.42 and $-1.5°$. Note that it is nonzero because of the lack of intensity information available in averaging unit vectors as first predicted by *Creer* [1983] and more recently discussed by *Khokhlov et al.* [2001] and by *Love and Constable* [2003]. The size of this bias in estimating the average field direction from unit vectors depends on the statistical properties of the PSV: we noted varying values depending on whether we used CJ98.GAD or either of the nonzonal versions of the PSV model (CJ98.nz.GAD) in which excess variance is concentrated in either the ${h_2}^1$ or ${g_2}^1$ spherical harmonic. We note that the average inclination anomaly predicted by the PSV model CJ98.GAD is compatible with the bounds seen for data from the southwestern U.S.A. given in Figures 15 and 16. Although the average inclination anomaly is consistent for the entire range of N_{site}, the standard error of the estimate, $\sigma_{\Delta I}$ from a single simulation decreases from a standard deviation of 4.4° for $N_{site} = 5$ to 0.3° for $N_{site} = 1000$, indicating that we can expect occasional large inclination anomalies for studies with small numbers of sites, even when these are not representative of time-averaged geomagnetic field behavior. With the variance due to paleosecular variation used in CJ98.GAD, about 300 sites would be required to detect the actual inclination bias due to averaging directional (rather than full vector) data and distinguish the bias from zero at the 95% confidence level. The problem of how to recover unbiased estimates of the average field is discussed further in the global modeling context by *Hatakeyama and Kono* [2002] and for local studies like the present one by *Love and Constable* [2003].

[58] In Figure 19b we see that the calculation of S_p, VGP dispersion about the rotation axis, also stabilizes to a consistent estimate for N_{site} somewhere between 100 and 1000. The issue is however complicated by the choice of the VGP cutoff value Θ_c that demarcates excursional or transitional behavior from "normal" secular variation, an issue discussed in more detail in the next section. Nevertheless we can state with some confidence that if random sampling in time is used without further information about temporal variations, then regional compilations of several hundred sites are desirable in order to adequately study TAF and PSV.

4.5. Should Transitional VGPs Be Excluded From TAF and PSV Dispersion Studies?

[59] In Figure 19a all VGPs deviating from the geographic axis by more than 45° were eliminated from the calculation. Conventional wisdom has it that the study of so-called "normal secular variation" should not include data with VGP directions that deviate so far from the geographic axis that they might be considered excursional or representative of a transitional field, and lava flow compilations for studying PSV have traditionally excluded low latitude VGPs. The definition of what constitutes a low latitude VGP has been somewhat variable, typically ranging from 45° to 55° in VGP latitude for a normal polarity field. Here we characterize the departure from normal secular variations using Θ, the angular deviation of the VGP direction from the geographic axis, according to the convention introduced by *Vandamme* [1994] and investigate the influence on PSV dispersion estimates of varying the cutoff value Θ_c. Thus for normal polarity data Θ is just the VGP colatitude θ, while for reverse polarity it is $180° - \theta$. *Vandamme* [1994] advocated selecting the cutoff value Θ_c to be dependent on latitude according to the relation

$$\Theta_c = 1.8S_p + 5°$$

on the grounds that variations in S_p with latitude should be considered in determining the normal range of secular variation for a stable polarity field. MM97 applied this formulism with the results that Θ_c ranges from about 25° at the equator to a

maximum of 45° at the poles with their data sets. For the southwestern U.S.A. they found values of about 35°. It is, however, interesting to note that at mid latitudes between 3 and 4% of the VGPs drawn from CJ98.GAD are more than 45° away from the pole, yet this is not an excursional or transitional field. Such large deviations from the pole generally occur more often at high latitudes [see *Constable and Johnson*, 1999, Figures 6(h) and 7(h)].

[60] In Figures 19b and 19c the effects of using Θ_c = 45° as a cutoff in Figure 19a are investigated by allowing it to vary from 35° to 180° (i.e. no cutoff at all) corresponding to the different symbols. We observe that when data with low VGP latitudes are rejected, S_p is systematically lower than when all the data are included. This bias is substantial for the statistical model CJ98.GAD, with S_p increasing from 12.7 to 18.7° as Θ_c ranges from 35° up to 180° for $N_{site} = 10^5$. If such an effect also occurs in the real data, then one must question whether it is justifiable to reject good quality data with Θ lying far from the geographic axis, especially if the geomagnetic field exhibits a smooth continuum of behavior as it moves towards what would be interpreted as excursional or transitional states.

[61] The large data set available for southwestern U.S.A. allows us to assess the influence that Θ_c selection has on the results (Figure 20). After eliminating sites with $k < 100$ and $N < 5$ we are left with 135 normal, 156 reverse or a combined polarity data set of 291 sites. Figure 20a presents the scalar average of the inclination anomaly as a function of Θ_c for the normal, reverse, and combined polarity data in a comparison with estimates from 100 distinct Monte Carlo simulations from CJ98.GAD each with 291 sites. We use the scalar average because of differences in the site locations; readers should note that this scalar average inevitably has a larger bias than the vector averages plotted in Figure 19a. The scalar inclination anomaly decreases systematically from zero to a negative anomaly of several degrees, as Θ_c increases and fewer data are eliminated from the average. The influence on S_p is much larger, with S_p ranging from less than 5° to more than 20° as Θ_c ranges from 5 to 90°. Of course no sensible person would argue for rejecting data with Θ_c = 5°, but the results presented in Figure 20 do confront the question of how to choose an appropriate Θ_c beyond which the field can be considered to have overstepped the bounds of normal secular variation and entered an excursional or transitional state. What we see is that while both S_p and $\Delta\bar{I}$ do level off to stable values these are really only achieved when essentially all the data are included: we might be justified in rejecting VGPs lying far from the geographic axis if these could be considered distinct from those representing normal secular variation, but no such feature is apparent in these curves. What we see is a continuum of possible values all of which could plausibly have been generated by normal secular variation. Any distinction on the basis of Θ_c is purely arbitrary. Truncated results cannot therefore be taken as representative of the full range of normal geomagnetic field behavior. This argument applies not just to the southwestern U.S.A. data, but also to the results of simulation from CJ98.GAD which readers may also note appears to substantially underestimate the variance in the real paleomagnetic data. While it is tempting to suspect that this is because CJ98.GAD was built on a truncated data set, the generally poor agreement even at low Θ_c suggests that the model is inadequate for other reasons. The long term goal of these TAF and PSV investigations is to produce a global model that is compatible with data sets like that generated for southwestern U.S.A., but that will not be attempted here.

[62] One might wonder why others (e.g., MM97) have apparently considered *Vandamme*'s [1994] method for selecting Θ_c perfectly adequate while our results would suggest otherwise. We suspect the flaw lies in it having been tested on data sets that exhibit smaller variances than the one here. MM97 describe a test using synthetic data generated from the IGRF 1965 field model for which the lowest VGP latitudes are greater than 50°. In such a circumstance it appears to perform better.

[63] Based on the CJ98.GAD simulations, which underestimate (rather than overestimate) the field variability for southwestern U.S.A. we expect to find an average inclination anomaly of at least

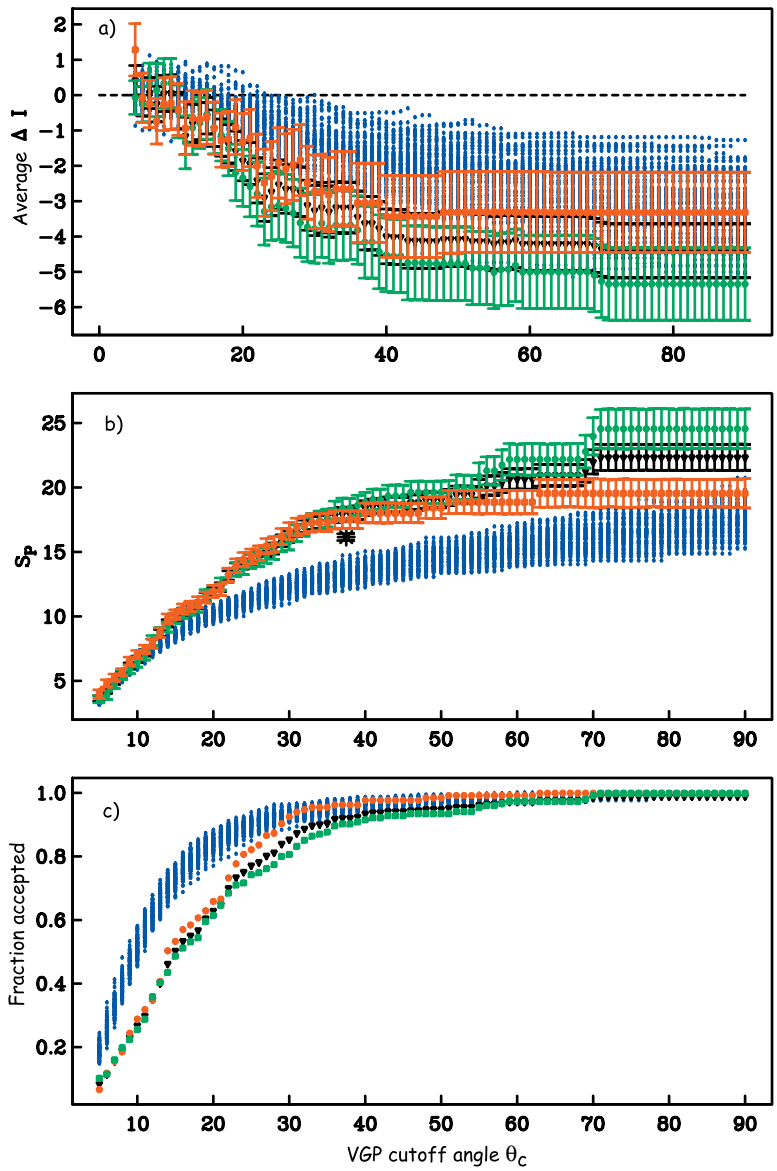


Figure 20. Influence of Θ_c on results from actual data and 100 simulations from CJ98.GAD each with 291 sites as for the combined normal and reverse data sets. Blue triangles represent simulations, red squares the normal polarity, green circles reverse polarity, and black triangles combined polarity results for the SWUS data set. Error bars are one standard error based on bootstrap estimates from the available data sets a) Scalar average of inclination anomalies, b) Dispersion about the rotation axis, S_p; black asterisk is MM97's global estimate of S_p for the 35–40° latitude band. (c) Fraction of data accepted.

−1.5° even in the absence of systematic non-axial-dipole field contributions. The large variance in inclination values make it is unlikely that this anomaly can be distinguished from zero with less than a few hundred sites. It appears that the larger N_{site} is the better and that if N_{site} is less than about 100, estimates of ΔI and S_p are likely to be quite unreliable.

[64] The foregoing discussion leads us to set Θ_c = 90° and include all available data with $N \geq 5$ and $k \geq 100$ in estimates for S_p. For the 22 sites meeting these criteria from the new data (shown as squares in Figure 13), $S_p = 15.0^{17.9}_{11.9}$. (The correction for within site scatter S_w is negligible.) S_p for the 135 normal sites meeting these criteria in the data set combining published and new data

is $19.6^{21.8}_{17.5}$. Our estimates for S_p for the 156 reverse data meeting the $k \geq 100$, $N \geq 5$ criteria with no VGP latitude cutoff is $24.6^{26.9}_{21.0}$. But others have not included data they considered not representative of normal secular variation and, using $\Theta_c = 35°$, MM97 estimate that S falls between 15.6 and 16.7 for the global latitude band 35–40°. Using the same cutoff value, we reject a total of 23 sites and retain $N_{site} = 130$ with $S_p = 17.5^{18.9}_{16.2}$ for the normal data and $N_{site} = 138$, $S_p = 17.8^{19.1}_{16.1}$ for reverse. The estimates for dispersion for the complete southwestern U.S.A. data set are therefore consistent with those derived from the new San Francisco Volcanics data alone as well as with the predictions of MM97. We can therefore conclude that the quality selection criteria $N \geq 5$, $k \geq 100$ are robust and that the published data meeting these criteria are similar in quality to the data presented here.

5. Conclusions

[65] ● We have presented data from 47 new paleomagnetic sites obtained from the San Francisco Volcanics. The NRM directions were essentially random. Demagnetization behavior in many sites was characteristic of the removal of an IRM induced by in some cases multiple lightning strikes. The current in one such strike was estimated to be approximately 300,000 Amps.

[66] ● We compiled all the published data from the southwestern region of the United States. In a departure from previous time averaged field studies, we included data from all igneous units regardless of mode of emplacement or quality factors such as number of samples or scatter of directions. We found that earlier time averaged field compilations had included data from strongly faulted regions. Exclusion of data from these disturbed areas significantly reduces the scatter in the data. There was no observable difference in the data set for different modes of emplacement (dikes, vents, flows, tuffs, etc.), however.

[67] ● The combined data set, excluding regions affected by tectonics, was highly scattered and elliptically distributed. Realizing that there are many sources of scatter, including lightning, and that most of these manifest themselves as highly scattered within site results, we proceeded to reject sites based on several measures of within site scatter. Rejection of sites with Fisher precision parameters k less than about 100 resulted in a distribution of directions that was axially symmetric. A similar effect can be achieved by using $\alpha_{95} = 5°$ as a cutoff value. Because α_{95} can be manipulated by simply using more samples without significantly improving the underlying dispersion of the site, we prefer the use of k as a selection criterion. Monte Carlo simulations suggest that at least five independently oriented samples are necessary to estimate k reliably, so that this should be used as an additional selection criterion in identifying high quality observations.

[68] ● Using simulations from CJ98.GAD, a statistical model for the paleosecular variation, we found that several hundred sites are necessary to accurately determine the TAF and its variance. Following earlier investigators we note that estimates of the TAF field are biased by the use of averages based on unit vectors rather than the full magnetic field vector. We can find no justification in the data for rejecting those data points with VGPs lying beyond any critical angular deviation, Θ_c, from the geographic axis. Such a procedure is likely to severely underestimate the field variance associated with PSV, although it might be argued that it reduces the bias in estimating the average direction. However, other more consistent techniques could do a better job. A preliminary method for simultaneously finding unbiased estimates of the average local field and its variance is discussed by *Love and Constable* [2003]. A generalization of this technique could be usefully applied to this regional data set.

[69] ● Using the criteria that all sites have at least five separately oriented samples and that the directions of the sites have a Fisher precision parameter of at least 100 we retain 291 sites from the region. Both the normal and reverse data sets are consistent with Fisher distributions and the average direction cannot be considered inconsistent with that expected from a geocentric axial dipole. However, we must emphasize that the expected direction for an axial dipole depends on the statistical properties

of the paleosecular variation which still need to be adequately characterized. When we combine our criteria for k and N with the rejection of VGPs for which $\Theta > 35°$ the dispersion of the associated VGPs is consistent with previously published global field compilations. While the consistency in results is satisfying, a more realistic estimate of dispersion includes all observations satisfying the quality criteria for k and N.

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